

Arjuna JEE 2027
Backlog Killer



Units and Measurement

(One Shot)

Subject : Physics

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Today's *Goal of the class*

1 Physical quantities and classification

2 Units and Dimensions

3 Application of dimensions

4 Errors analysis

5 Significant Figures



Physical Quantity

Any Qty which Can be Measured
using Physical Instruments.

S.No	Physical Quantities	Instruments
1.	length	Scale
2.	Mass	Weighing Machine
3.	Time	Clock
4.	Current	Ammeter
5.	Voltage	Voltmeter.

Classification of Physical Quantity

on the basis of direction

Scalar

Mag = ✓
but do not have
any direction &
follows Scalar add.

Ex: - Mass, Length, Time
* Current, Temp, dist, Speed.

↓
Scalar

Vector

Mag = ✓
Direction = ✓
& follows Vector laws
of addition.

Ex: - Disp, Velocity, acc
Force, Momentum.

on the basis of dependency

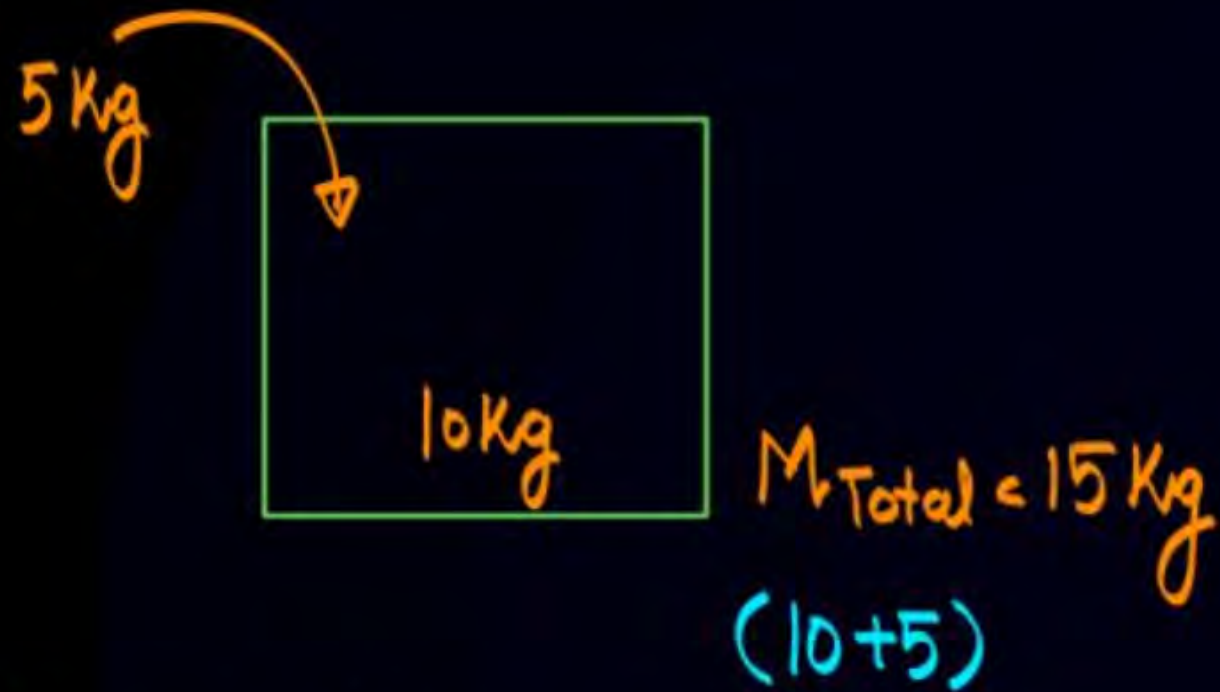
fundamental
7 Nos.

1. Mass
2. Length
3. Time
4. Current
5. Temperature
6. Amount of Subs.
7. Luminous Intensity

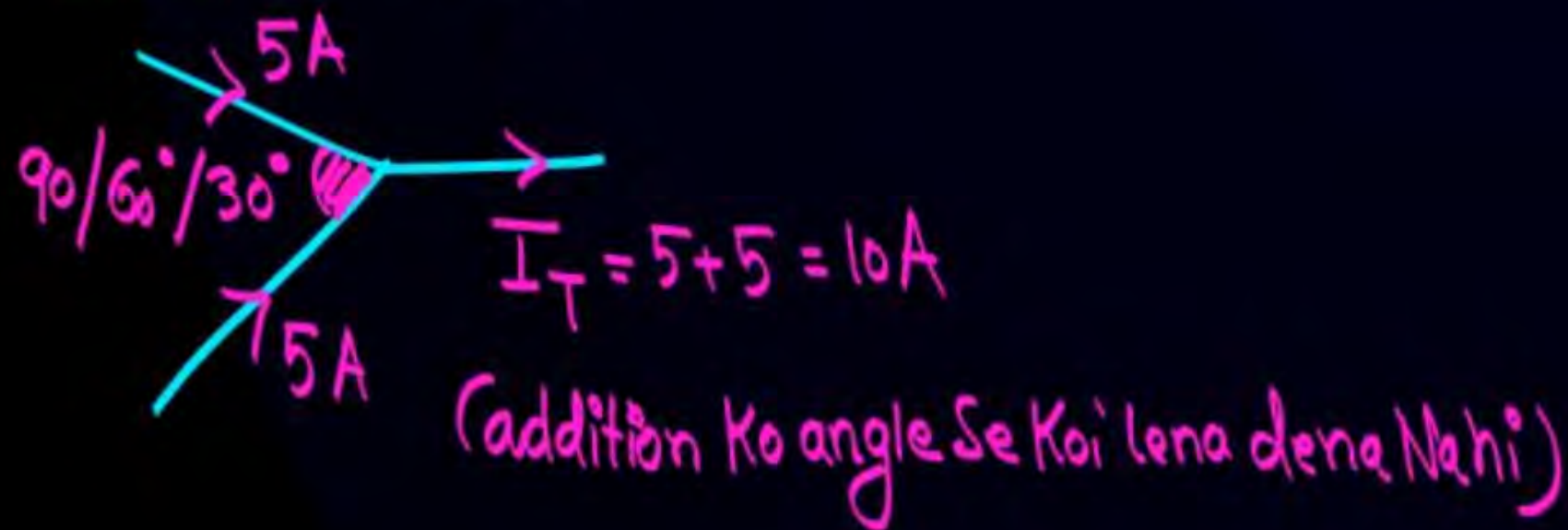
Derived

→ Speed, Velocity,
acc, Force, work
Power, Momentum,
density.

Mass $\Rightarrow$ "Scalar"

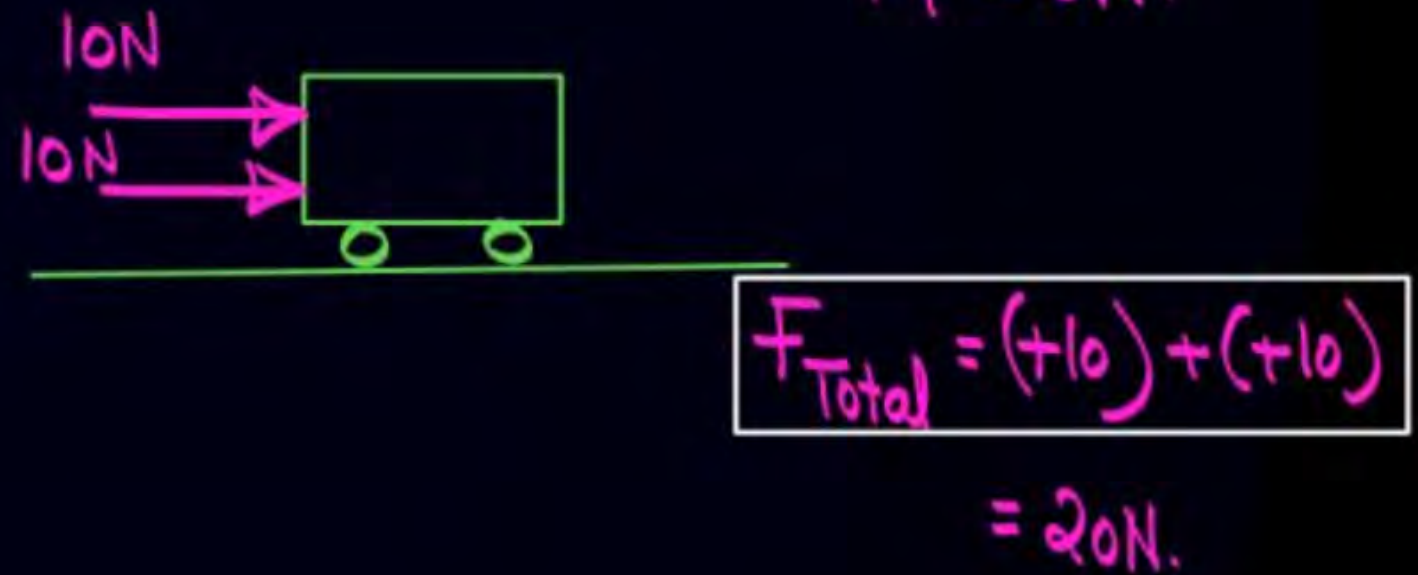
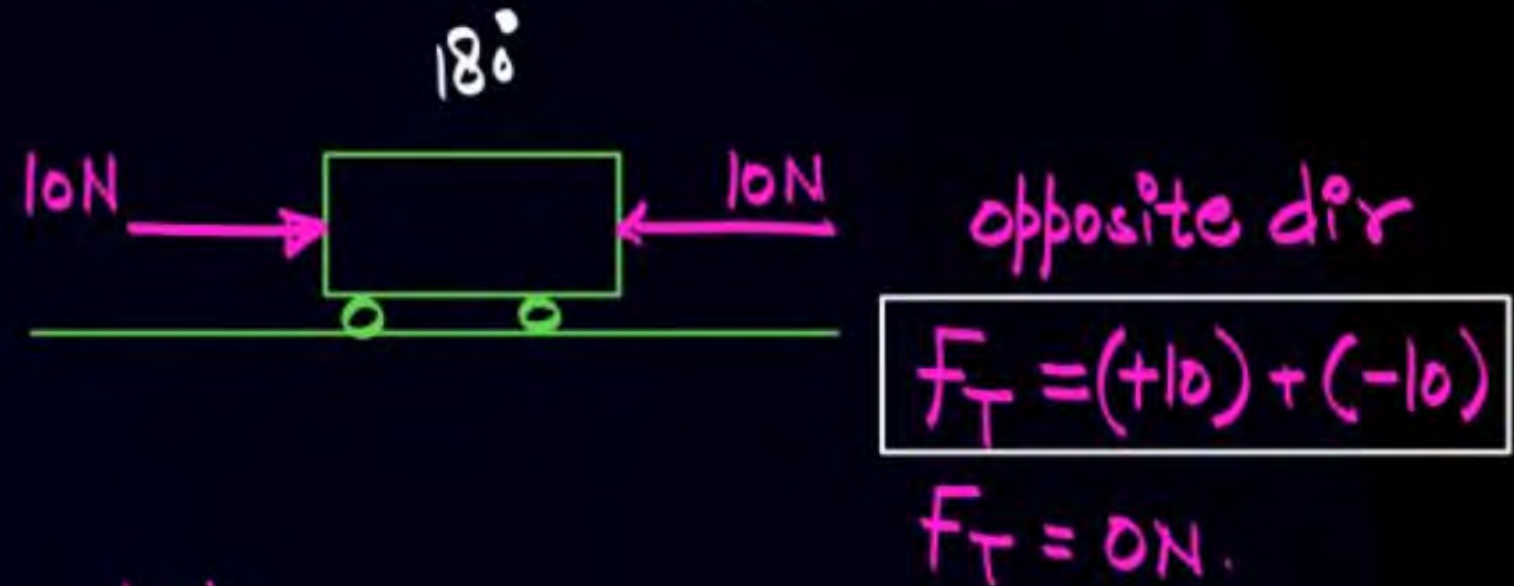


Current: - "Scalar"



Force

$\therefore$ = Whenever they are added we check the Angle & then add.





Units

Standard Reference Value of any PQ $\rightarrow$ unit.

- $\rightarrow$ Mass $\rightarrow$ Kg
- $\rightarrow$ Length $\rightarrow$ m
- $\rightarrow$ Time $\rightarrow$ s
- $\rightarrow$ Current $\rightarrow$ A
- $\rightarrow$ Temp $\rightarrow$ Kelvin
- $\rightarrow$ Amount of Sub $\rightarrow$ Moles
- $\rightarrow$ Luminous Intensity $\rightarrow$ Candela.

Characteristics of Units:

- (i) Well defined
- (ii) Invariability (constancy)
- (iii) Reproducibility
- (iv) Convenience in use





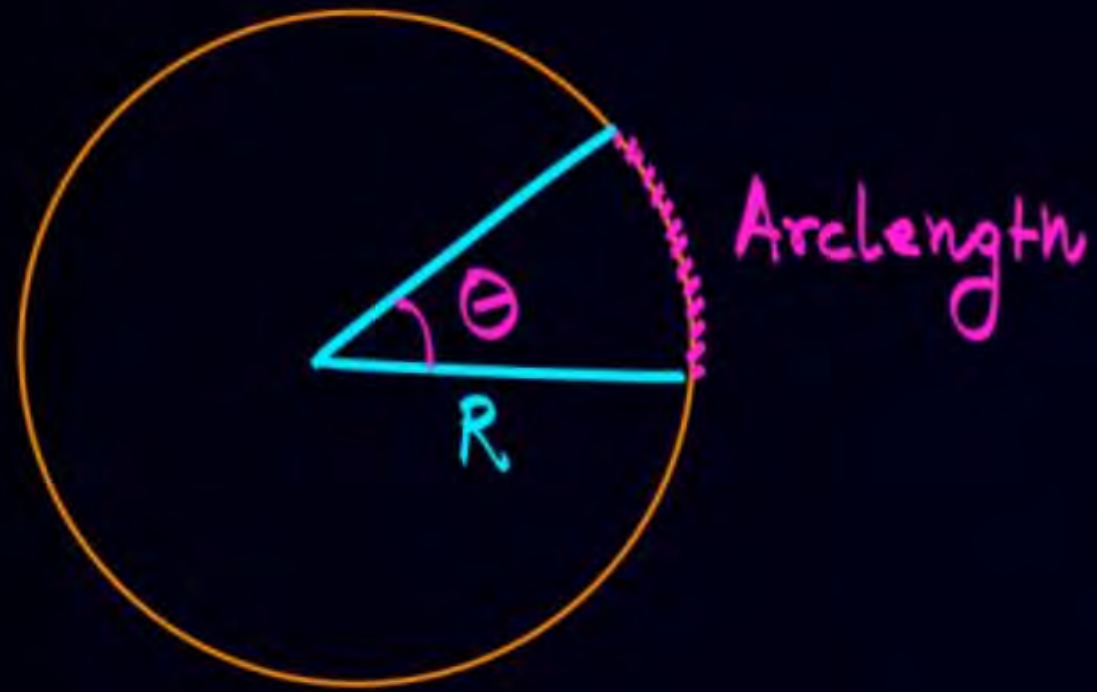
System of Units

Physical Quantity	F PS	CGS	MKS
Length	Foot	Centimeter	meter
Mass	Pounds	gram	Kilogram
Time	Seconds	Seconds	Seconds

International systems of units: (SI unit system) $\Rightarrow$ MKS + 2 supplementary units.
(Angle, Solid Angle).

Supplementary Units:

Plane Angle: " θ "



$$\textcircled{1} \theta = \frac{\text{Arc length}}{\text{Radius}}$$

$$\textcircled{2} \text{unit} = \text{unitless} \times \boxed{\text{Radians}}$$

$$\textcircled{3} \boxed{180^\circ = \pi \text{ radians}}$$

other units

→ degrees ($^\circ$)
→ Minutes ($'$)
→ Seconds ($''$)

Doubt:-

6 seconds → Time

6' Angle → Angle.

#



60 div, Each div = Minute. $\boxed{1^\circ = 60'}$

if Each Minute is divided 60 Equal parts $\boxed{1' = 60''}$
Each div = Second.

QUESTION 1

Convert the following angles to radians:

$$1. 30^\circ \rightarrow 30 \times \frac{1^\circ}{180} = 30 \times \frac{\pi}{180} = \frac{\pi}{6} \text{ rad.}$$

$$2. 45^\circ \rightarrow 45 \times \frac{\pi}{180} = \frac{\pi}{4} \text{ rad}$$

$$3. 60^\circ \rightarrow 60 \times \frac{\pi}{180} = \frac{\pi}{3} \text{ rad}$$

$$4. 90^\circ \rightarrow 90 \times \frac{\pi}{180} = \frac{\pi}{2} \text{ rad}$$

$$5. 120^\circ \rightarrow 120 \times \frac{\pi}{180} = \frac{2\pi}{3} \text{ rad}$$

$$6. 180^\circ \rightarrow 180 \times \frac{\pi}{180} = \pi \text{ rad.}$$

$$7. 270^\circ \rightarrow 270 \times \frac{\pi}{180} = \frac{3\pi}{2} \text{ rad.}$$

$$180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad.}$$

QUESTION 2

Convert the following angles to degrees:

$$1. \frac{3\pi}{4} \Rightarrow \frac{3\cancel{\pi}}{\cancel{4}} \times \frac{180}{\cancel{\pi}} = 135^\circ$$

$$2. \frac{\pi}{6} \Rightarrow \frac{\pi}{6} \times \frac{180}{\pi} = 30^\circ$$

$$3. \frac{2\pi}{5} \Rightarrow \frac{2\pi}{5} \times \frac{180}{\pi} = \frac{360}{5} = 72^\circ$$

$$4. \frac{3\pi}{2} \Rightarrow \frac{3\pi}{2} \times \frac{180}{\pi} = 270^\circ$$

$$180^\circ = \pi \text{ rad}$$

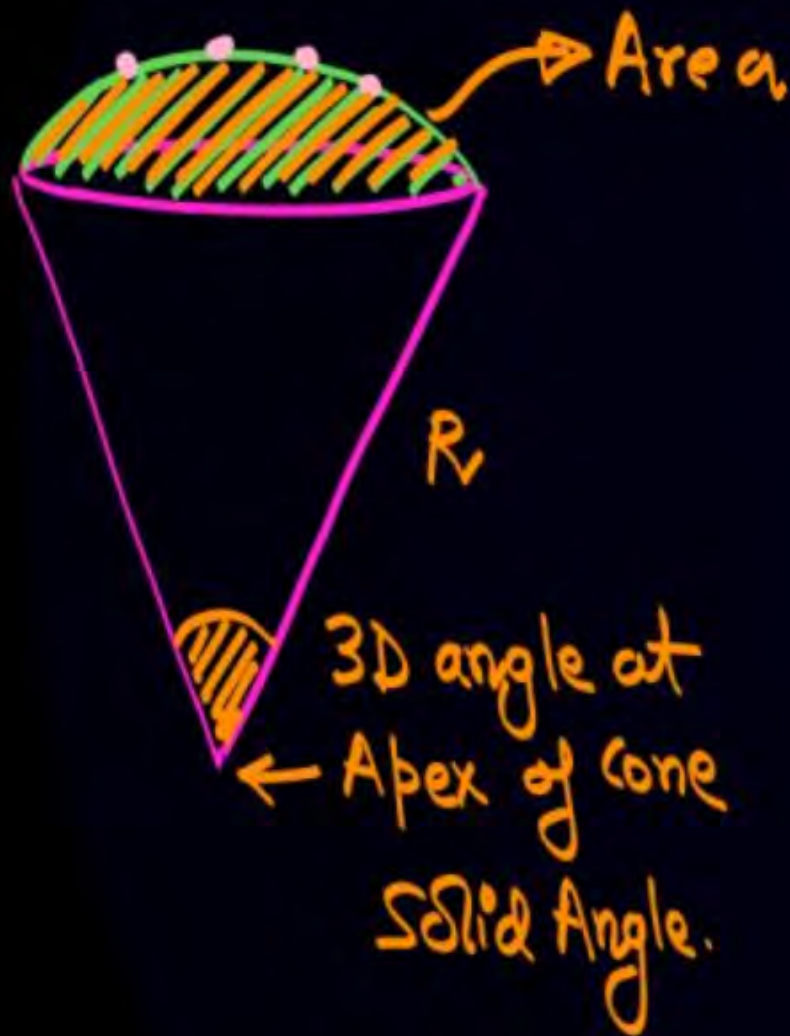
$$\frac{180}{\pi} = 1 \text{ rad}$$

Supplementary Units:

Solid Angle:

$$\odot \Omega = \frac{\text{Area}}{R^2}$$

$\odot$ unit $\rightarrow$ Steradians



#

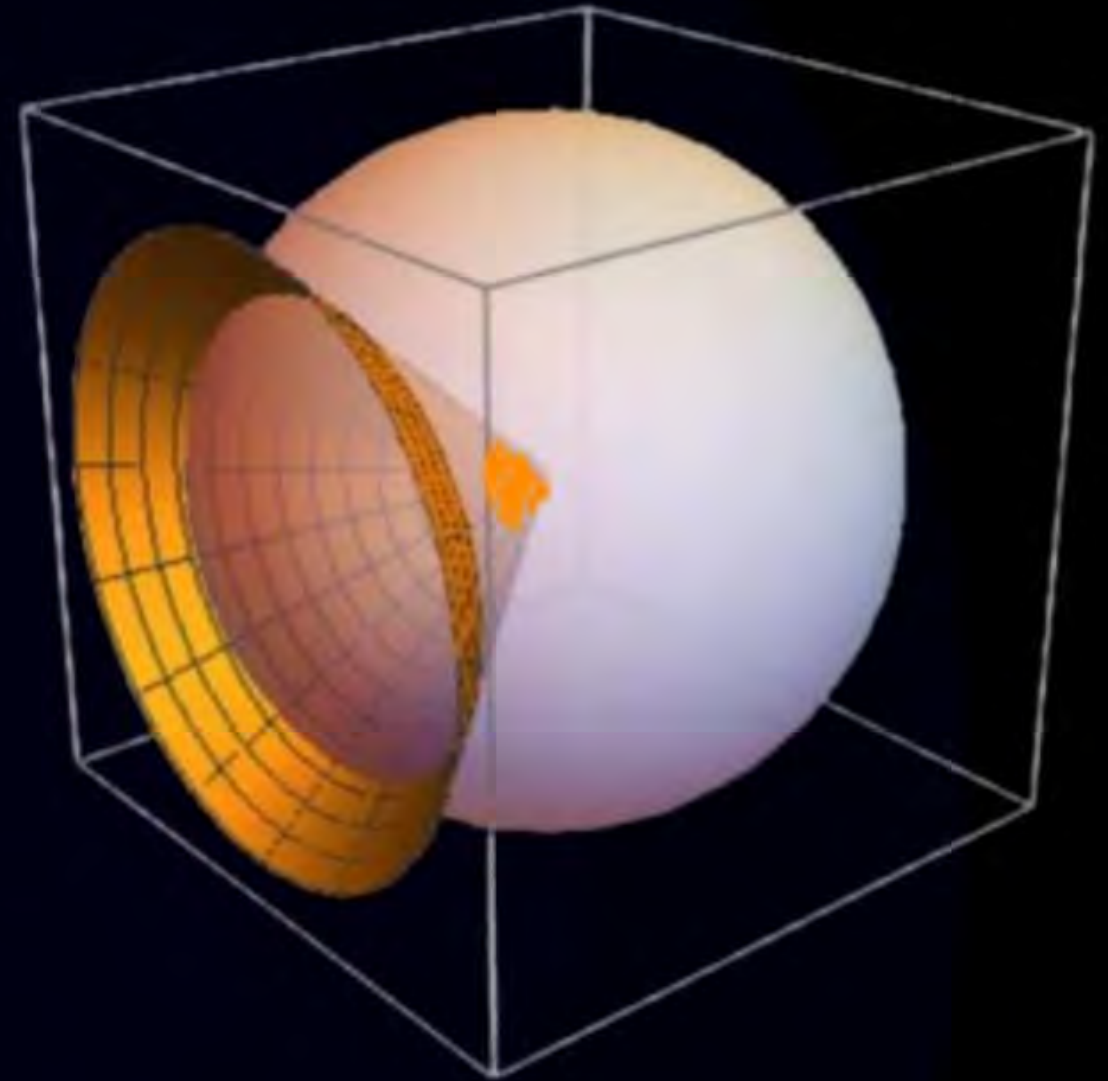


$$\Omega = \frac{2\pi R^2}{R^2} = 2\pi \text{ Steradians}$$

#



$$\Omega = \frac{4\pi R^2}{R^2} = 4\pi \text{ Steradians.}$$





Important terms

$$\text{Light Speed} = 3 \times 10^8 \frac{\text{m}}{\text{s}}$$



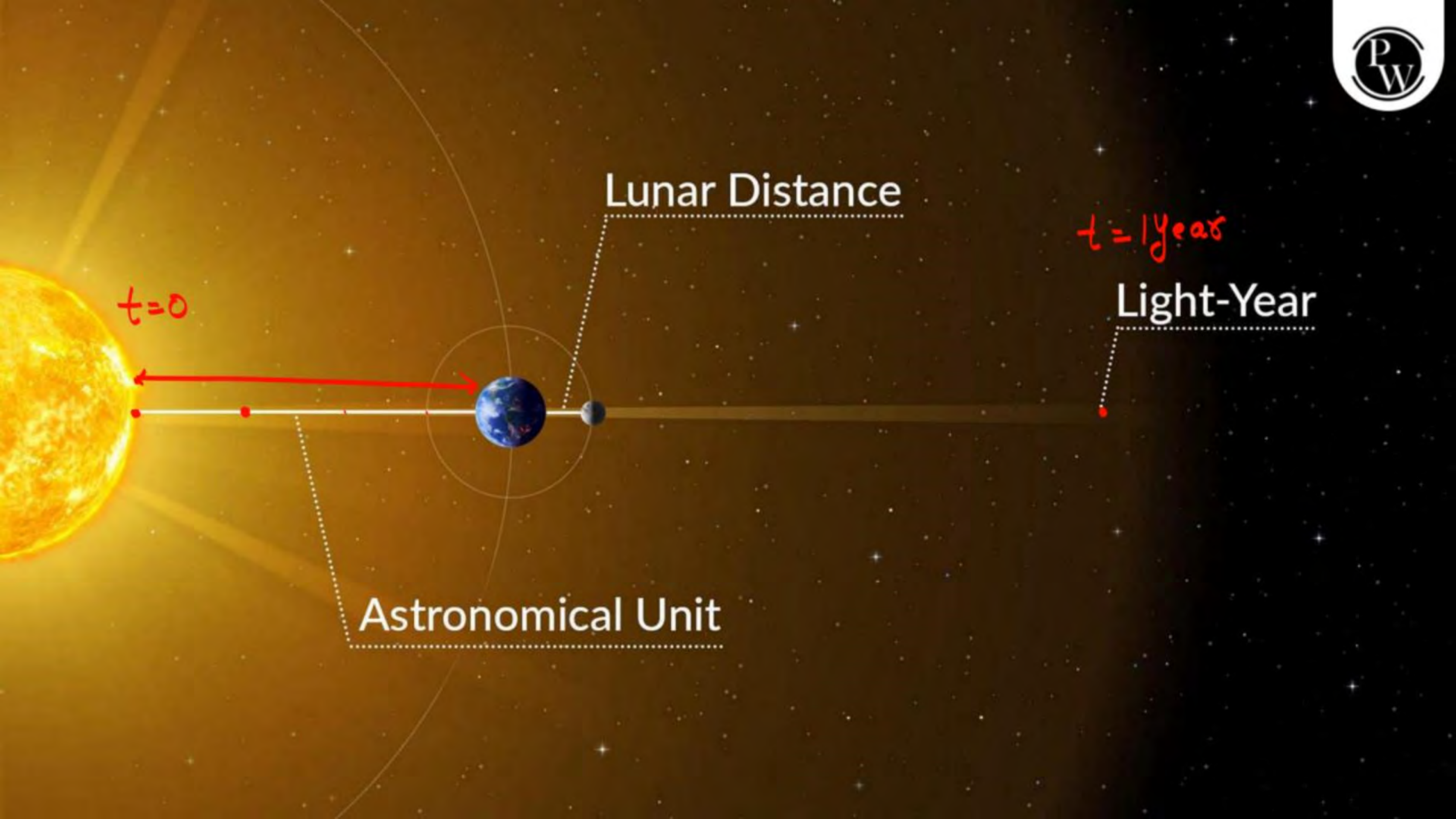
Light Year: (9.5×10^{15} m) $\Rightarrow$ distance travelled by light in complete one year.

Astronomical Unit: (1.5×10^{11} m) distance between Sun & Earth.

Parsec: (3.0857×10^{16} m)

Radius of Arc whose Arc length = 1 AU
& angle at Centre is 1".





Lunar Distance

$t=1\text{year}$

Light-Year

Astronomical Unit



Dimensions

Powers to which fundamental PQ are raised to Represent a PQ.

Fundamental (7)

1. Mass $\longrightarrow M$
2. Length $\longrightarrow L$
3. Time $\longrightarrow T$
4. Current $\longrightarrow A \text{ or } I$
5. Temperature $\longrightarrow K$
6. Amount of Subs. $\longrightarrow \text{Mol}$
7. Luminous Intensity $\longrightarrow \text{Cd}$.

Angle = $M^0 L^0 T^0$, radians

Solid Angle = $M^0 L^0 T^0$, steradians

Derived PQ

$$\text{Velocity} = \frac{\text{disp}}{\text{time}} = \frac{L}{T} = [L^1 T^{-1}]$$

$$\text{acc} = \frac{\text{vel}}{\text{Time}} = \frac{L T^{-1}}{T} = [L^1 T^{-2}]$$

$$\text{Force} = \text{mass} \times \text{acc} = [M^1 L^1 T^{-2}]$$

$$\text{Work} = \text{Force} \times \text{dist} = M L T^{-2} \times L = [M^1 L^2 T^{-2}]$$

$$\text{Momentum} = m \times v = [M^1 L^1 T^{-1}]$$

$$\text{Power} = \text{work/time} = [M^1 L^2 T^{-3}]$$

$$\text{Density} = \frac{\text{Mass}}{\text{Vol}} = [M^1 L^{-3}]$$

YE Revise kar liya toh Life Jhingalala

- All length (Radius of Gyration, Astronomical unit, parsec, light year) has dimension $[L]$
- All velocities have dimension $[LT^{-1}]$ but angular velocity has dimension $[T^{-1}]$
- All Acceleration has dimensions $[LT^{-2}]$ but angular acceleration has dimension $[T^{-2}]$
- All Force (Thrust, viscous drag, normal reaction, tension, weight) has dimensions $[MLT^{-2}]$ but surface tension Spring constant has dimension $[MT^{-2}]$
- All Work, Energies, Torque (Couple, Moment of force) has dimensions $[ML^2T^{-2}]$
- $\rightarrow \frac{F}{A}$ Pressure, stress has dimension $[ML^{-1}T^{-2}]$, All modulus (Young, Bulk, Shear, Modulus of rigidity) terms has dimensions of Pressure
- All gradient terms are derived by dividing by Length
 - $\rightarrow \text{force gradient} = \frac{F}{l} = [MT^{-2}]$
 - $\rightarrow \text{Pressure gradient} = \frac{P}{l} = [ML^{-2}T^{-2}]$
- Momentum / Impulse has same dimensions of $[MLT^{-1}]$
- Angular momentum is Momentum $\times$ distance with dimensions $[ML^2T^{-1}]$

- Dimensions of **Linear** density $\left(\frac{M}{L}\right) = [ML^{-1}]$, **Areal** Density $\left(\frac{M}{A}\right) = [ML^{-2}]$, **Volumetric Density** $\frac{M}{V} [ML^{-3}]$
 - Dimensions of **Linear Charge** density $\frac{[AT]}{L}$, **Areal** charge Density $\frac{AT}{L^2}$, **Volumetric** charge Density $\frac{[AT]}{L^3}$
 - Gravitational Constant is derived from formula $F = \frac{Gm^2}{r^2}$
 - Planck's Constant is derived from formula $E = h\nu$
 - Resistance is derived from formula $Power = I^2 R$
 - Capacitance is related to resistance by relation $RC = Time$
 - Inductance (L) is related to resistance by formula $\frac{L}{R} = Time$
 - Potential/ Electromotive Force is derived from formula $P = VI$
 $\xrightarrow{\text{Potential}}$
 - Permittivity dimensions is calculated from formula $F = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2}$
 - Coefficient of viscosity is calculated from formula $F = \eta r v$
- Density $1. \frac{M}{V} = \frac{ML}{L^3}$
 Charge density $\frac{Q}{Vol} = \frac{AT}{L^3}$
 Energy density $\frac{Energy}{Volume} = \frac{ML^2T^{-2}}{L^3}$
 $I = \frac{Charge}{time}$
 $AT = Charge$
- Planck's Cons
 Energy = $h\nu$
 freq = $\frac{1}{T}$

- Permeability is calculated from formula $\text{speed} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$
- Angle and Solid Angle has dimensions $M^0 L^0 T^0$ and Units radians, Steradians

- Boltzmann constant dimensions is derived from formula $\text{Energy} = K_B (\text{Temp})$

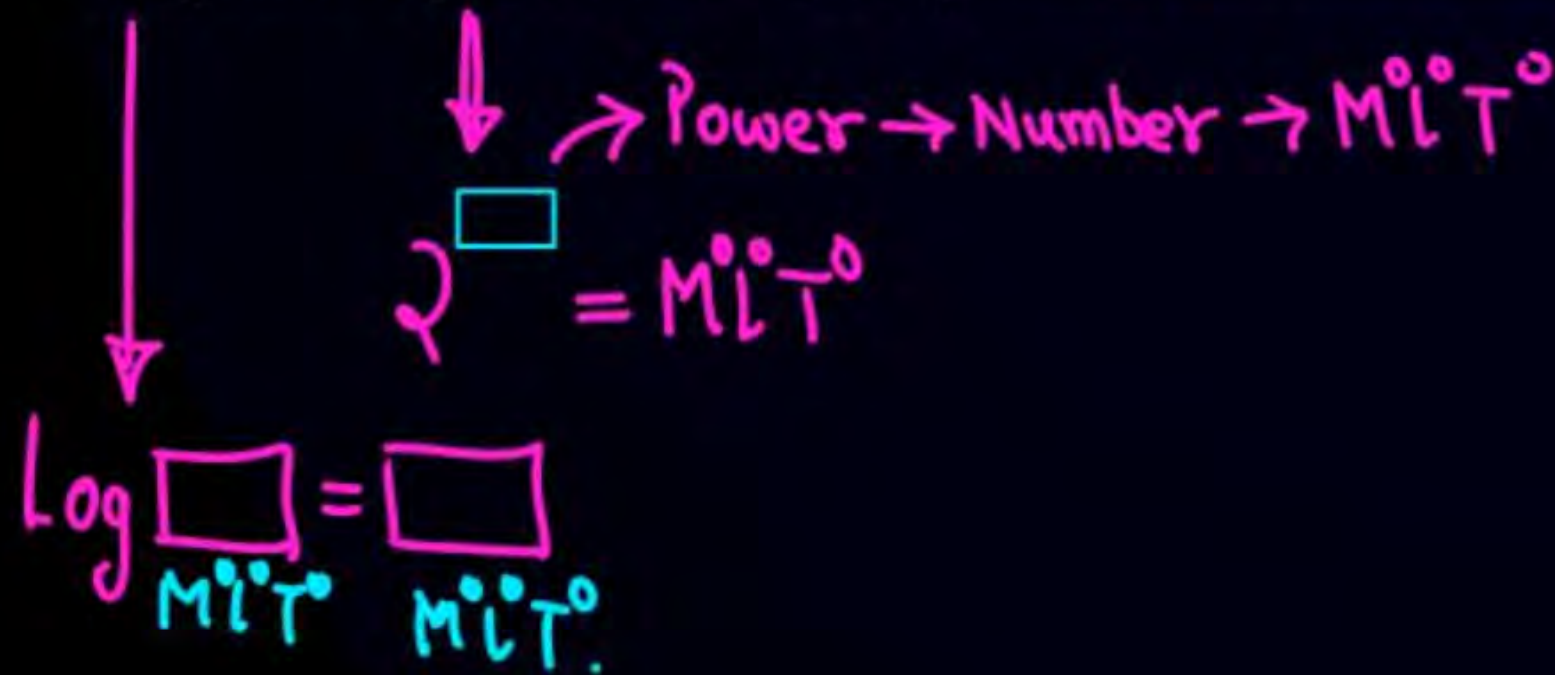
- Electric Field dimensions is derived from formula $F = qE$

- Magnetic Field dimension is derived from formula $E = Bc$ $c = \text{speed} = LT^{-1}$

$\sin \theta, \cos \theta, \tan \theta, \cot \theta, \operatorname{cosec} \theta, \sec \theta = M^0 L^0 T^0$

- Trigonometric ratio are always $M^0 L^0 T^0$

- Log and exponents are always $M^0 L^0 T^0$





Principle of Homogeneity

Same PQ Ke beech mein (+, -, =, >, <, ≥, ≤) Laga Sakte ho.

Length + Area
→ $\underline{10m} + \underline{2m^2} = X$

→ $5^\circ C + 2A = X$

→ $1cm + 5mm = 1cm + 0.5cm = 1.5cm$

Length + Length = Length

→ $10m - 10m = 0m$
length - length length

→ $1m = 100cm$

$$L + L = L$$

$$L - L = L$$

$$L = L$$

Principle of Homogeneity

QUESTION 3

If a and b are two physical quantities having different dimensions then which of the following can be logical?

~~A~~ $a + b$ ✗

~~B~~ $a - b$ ✗

☒ C a/b ✓

~~D~~ $e^{a/b} = 2.7^{(a/b)}$

$e = \text{Exponent} = (2.7)$

$[a] = \text{Charge}$
 $[b] = \text{time}$

$\frac{a}{b} \neq M^0 L^0 T^0$

→ Kisi bhi number ki Power
 $= [M^0 L^0 T^0]$

We Can Multiply & divide
any two PQ.

→ $\text{Speed} = \frac{\text{dist}}{\text{time}}$

→ $\text{Force} = \text{mass} \times \text{acc}$

→ $I = \frac{\text{Charge}}{\text{time}}$

→ $\text{Power} = \text{Vol} \times \text{Current}$

$$4) \boxed{x} = \boxed{\dot{a}} + \boxed{\dot{b}t} + \boxed{\dot{c}t^2} + \boxed{\dot{d}t^3};$$

$\downarrow$
[L]

$$[a] = L$$

$$bT = L \Rightarrow b = [LT^{-1}]$$

$$cT^2 = L \Rightarrow c = [LT^{-2}]$$

$$dT^3 = L \Rightarrow d = [LT^{-3}]$$

$t = \text{time}$.

$$5) \boxed{F} = \boxed{\dot{a}x^2} + \boxed{\dot{b}t^{1/2}}$$

$$F = MLT^{-2}$$

$$x = L$$

$$t = T$$

$$aL^2 = MLT^{-2}$$

$$a = [ML^{-1}T^{-2}]$$

$$bT^{1/2} = MLT^{-2}$$

$$b = \frac{MLT^{-2}}{T^{1/2}} = MLT^{-2} \times T^{-1/2} = [MLT^{-5/2}]$$

$$6) \overset{L}{x} = \overset{L}{a} + \overset{L}{\frac{b}{a}t} + \overset{L}{\frac{ct^2}{d-P}} + \overset{L}{et^3}$$

$x = \text{length} = [L]$

$t = \text{time} = [T]$

$P = \text{Pressure} = [ML^{-1}T^{-2}]$

$$et^3 = L$$

$$e = [LT^{-3}]$$

$$\# [d] = [P]$$

$$d = [ML^{-1}T^{-2}]$$

$$a = [L]$$

$$\frac{b}{a}T = L$$

$$\frac{b}{L}T = L$$

$$b = [L^2T^{-1}]$$

$$\# \frac{CT^2}{ML^{-1}T^{-2}} = L$$

$$C = \frac{MT^{-2}}{T^2} = [MT^{-4}]$$

$$7) E = \frac{B-x^2}{At}$$

$E = \text{Energy} = ML^2T^{-2}$

$x = \text{length} = L$

$t = \text{Time} = T$

$$[B] = L^2$$

$$E = \frac{\overset{L^2}{B} - \overset{L^2}{x^2}}{AT}$$

$$ML^2T^{-2} = \frac{L^2}{AT}$$

$$A = \frac{1}{MT^{-3}T}$$

$$A = M^{-1}T$$

8) $\text{Force} = \frac{\text{Force} \cdot X}{\sqrt{\text{Density}}} + \text{Force} \cdot Y$

$\downarrow$ $\downarrow$

$[MLT^{-2}]$ $\frac{\text{Mass}}{\text{Vol.}} = [ML^{-3}]$

$$Y = [MLT^{-2}]$$

$$\frac{X}{(ML^{-3})^{1/2}} = MLT^{-2}$$

$$X = MLT^{-2} \times M^{1/2} L^{-3/2}$$

$$X = M^{3/2} L^{-1/2} T^{-2}$$

9) $F = \frac{\alpha}{\beta + \sqrt{d}}$

$$f = \text{force} = MLT^{-2}$$

$$d = \text{density} = ML^{-3}$$

$$F = \frac{\alpha}{\sqrt{\text{density}}}$$

$$MLT^{-2} = \frac{\alpha}{M^{1/2} L^{-3/2}}$$

$$MLT^{-2} \times M^{1/2} L^{-3/2} = \alpha$$

$$M^{3/2} L^{-1/2} T^{-2} = \alpha$$

$$\beta = [\sqrt{\text{density}}]$$

$$\beta = [(ML^{-3})^{1/2}]$$

$$\beta = M^{1/2} L^{-3/2}$$

$$10) \left(P + \frac{a}{V^2} \right) (V - b)$$

Pressure Volume Volume

$$P = \text{Pressure} = ML^{-1}T^{-2}$$

$$V = \text{Volume} = L^3$$

$$\frac{a}{(L^3)^2} = ML^{-1}T^{-2}$$

$$[b] = L^3$$

$$\frac{a}{L^6} = ML^{-1}T^{-2}$$

$$a = ML^5T^{-2}$$

$$11) P = b e^{(-at^2)} = b (2.7)^{-at^2}$$

$$P = \text{pressure} = ML^{-1}T^{-2}$$

$$t = \text{time} = T$$

$$P = \frac{b}{2.7^{at^2}}$$

$$M^0L^0T^0 \leftarrow \boxed{2.7^{at^2}} \rightarrow M^0L^0T^0$$

$$aT^2 = M^0L^0T^0$$

$$aT^2 = 1$$

$$a = T^{-2}$$

$$[b] = [P]$$

$$b = [ML^{-1}T^{-2}]$$



$$a, v_0 = ?$$

$$12) x = \frac{v_0}{a} (1 - e^{-at})$$

$$x = \text{length} = [L]$$

$$t = \text{time} = [T]$$

$$x = \frac{v_0}{a} \left[1 - \frac{1}{2.7^{at}} \right]$$

No. No.

$M^0 L^1 T^{-1}$

$$aT = M^0 L^1 T^{-1}$$

$$aT = 1$$

$$a = T^{-1}$$

$$[L] = \frac{v_0}{T^{-1}}$$

$$v_0 = [L T^{-1}]$$

$$\sin \frac{\text{Angle}}{M^0 L^1 T^{-1}} = M^0 L^1 T^{-1}$$

$$13) y = A \sin B \left(\frac{x}{v} - C \right)$$

$$y = \text{length}$$

$$x = \text{length}$$

$$v = \text{velocity}$$

$$A, B, C = \text{dim?}$$

$$y = A$$

$$\sin \left(\frac{Bx}{v} - B.C \right)$$

$$\text{length}$$

$$\text{length}$$

$$M^0 L^1 T^{-1}$$

$$M^0 L^1 T^{-1}$$

$$A = [L]$$

$$\frac{B L}{L T^{-1}} = M^0 L^1 T^{-1}$$

$$B.C = M^0 L^1 T^{-1}$$

$$B = T^{-1}$$

$$T^{-1} \times C = 1$$

$$C = \frac{1}{T^{-1}} = T$$

$$C = T$$

$$14) P = \frac{\alpha}{\beta} e^{-\frac{\alpha}{10}}$$

$$P = \text{pressure} = [ML^{-1}T^{-2}]$$

$$P = \frac{\alpha}{\beta} (2.7)^{-\frac{\alpha}{10}}$$

$$P = \frac{\alpha}{\beta} \times \frac{1}{(2.7)^{\frac{\alpha}{10}}} \quad M^0 L^0 T^0$$

$$\frac{\alpha}{10} = M^0 L^0 T^0$$

$$\alpha = M^0 L^0 T^0$$

$$[P] = \left[\frac{\alpha}{\beta} \right]$$

$$ML^{-1}T^{-2} = \frac{M^0 L^0 T^0}{\beta} \Rightarrow \beta = M^{-1} L T^2$$

$$\alpha, B = \text{dim} = ?$$

$$15) v = \frac{Ax}{B+x^2} + ct.$$

$$v = \text{velocity} = LT^{-1}$$

$$x = \text{length} = L$$

$$t = \text{time} = T$$

$$\frac{Ax}{B+x^2} = LT^{-1}$$

$$\frac{AK}{L^2} = LT^{-1}$$

$$A = L^2 T^{-1}$$

$$cT = LT^{-1}$$

$$c = LT^{-2}$$

$$B = L^2$$

$$16) \boxed{x} = \boxed{Ax^2} + \boxed{\frac{B}{t^2}} + \boxed{\frac{C}{x}}$$

$x = \text{length} = L$

$t = \text{time} = T$

$$Ax^2 = L$$

$$AL^2 = L$$

$$\boxed{A = [L^{-1}]}$$

$$\frac{B}{T^2} = L$$

$$\boxed{B = LT^2}$$

$$\frac{C}{L} = L$$

$$\boxed{C = L^2}$$

$$17) \boxed{F} = \boxed{\overset{\text{Force}}{A} \overset{\text{Force}}{v}} + \boxed{\overset{\text{Force}}{\frac{B}{t^2}}}$$

$$F = \text{force} = MLT^{-2}$$

$$v = \text{velocity} = LT^{-1}$$

$$t = \text{time} = T$$

$$A [LT^{-1}] = MLT^{-2}$$

$$\boxed{A = MT^{-1}}$$

$$\frac{B}{T^2} = MLT^{-2}$$

$$\boxed{B = ML}$$



$$18) \quad \alpha = \frac{F}{v^2} \sin(\beta t) \quad \alpha, v, \beta = ?$$

$M^0 L^1 T^0$ (above $\sin$)
 $M^0 L^1 T^0$ (below $\sin$)

$$F = \text{force} = M L T^{-2}$$

$$t = \text{time} = T$$

$$v = \text{velocity} = L T^{-1}$$

$$\beta T = M^0 L^1 T^0$$

$$\beta = T^{-1}$$

$$[\alpha] = \left[\frac{F}{v^2} \right]$$

$$\alpha = \frac{M L T^{-2}}{L^2 T^{-2}}$$

$$\alpha = [M L^{-1}]$$

QUESTION 19

$y = A \sin \left(\frac{B}{t^2} + Cv \right)$, here t is time and v is velocity and y is position. Then dimension of A.B.C depends on

A Time and length alone

B Mass, length alone

C Time alone

D Time and length alone

$$y = A \sin \left(\frac{B}{t^2} + Cv \right)$$

Handwritten notes for dimensions:

- $[y] = [A]$
- $A = [L]$
- $\frac{B}{t^2} = M^0 L^0 T^0$
- $Cv = M^0 L^0 T^0$
- $C L T^{-1} = 1$
- $C = L^{-1} T$

$$A \cdot B \cdot C = L \times T^2 \times L^{-1} T = T^3$$

$$\frac{B}{T^2} = M^0 L^0 T^0$$

$$B = T^2$$

$$Cv = M^0 L^0 T^0$$

$$C L T^{-1} = 1$$

$$C = L^{-1} T$$

QUESTION 20

$$F = ma$$

In a book, the answer for a particular question is expressed as $b = \frac{ma}{k} \sqrt{1 + \frac{2kl}{ma}}$ here m represents mass, a represents accelerations, l represents length. The unit of b should be

$$b = \frac{ma}{k} \sqrt{1 + \frac{2kl}{ma}}$$

$M^0 L^0 T^0$

$$\left[\frac{2kl}{ma} \right] = M^0 L^0 T^0$$

$$\frac{KL}{MT^{-2}} = 1$$

$$K = MT^{-2}$$

$$[b] = \left[\frac{MLT^{-2}}{MT^{-2}} \right] = [L]$$

Am/s ✗**B**m/s² ✗**C** ✓meter**D**

/sec

QUESTION 21

An unknown quantity ' α ' is expressed as $\alpha = \frac{2ma}{\beta} \log \left(1 + 2\beta \frac{l}{ma} \right)$, where m = mass, a = acceleration, l = length. The unit of α should be

A s^{-1} **B** m/s **C** m/s^2 **D**

meter

$$\alpha = \frac{2ma}{\beta} \log \left(1 + \frac{2\beta l}{ma} \right)$$

$M^0 L^0 T^0$

$M^0 L^0 T^0$

$$[\alpha] = \left[\frac{MLT^{-2}}{\beta} \right]$$

$$\alpha = \frac{MLT^{-2}}{MT^{-2}} = L$$

$$\left[\frac{2\beta l}{ma} \right] = M^0 L^0 T^0$$

$$\frac{\beta L}{MLT^{-2}} = 1$$

$$\beta = MT^{-2}$$

QUESTION 22

In the relation Pressure $P = \frac{\alpha}{\beta} e^{-\frac{\alpha z}{k\theta}}$, P is pressure, Z is distance, k is Boltzmann constant and θ is the temperature. The dimensional formula of β will be

- ☒ **A** $[M^0 L^2 T^0]$
- ☐ **B** $[ML^2 T]$
- ☐ **C** $[ML^0 T^{-1}]$
- ☐ **D** $[M^0 L^{-2} T^{-1}]$

$$P = \frac{\alpha}{\beta} e^{-\frac{\alpha z}{k\theta}}$$

$M^0 L^0 T^0$

$$P = \frac{\alpha}{\beta} \times \frac{1}{(2.7)^{\frac{\alpha z}{k\theta}}}$$

$M^0 L^0 T^0$

$[P] = \left[\frac{\alpha}{\beta} \right]$

$\cancel{ML^{-1}T^{-2}} = \cancel{MLT^{-1}} \times$

$\beta = L^2$

$\frac{\alpha k}{ML^2 T^{-2}} = M^0 L^0 T^0$

$\alpha = MLT^{-2}$

Energy = K_B (Temp)

$ML^2 T^{-2} = K_B \theta$

QUESTION 23

$$K(\text{Temp}) = \text{Energy}$$

The work done by a gas molecule in an isolated system is given by, $W = \alpha \beta^2 e^{-\frac{x^2}{\alpha kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature, α and β are constants. Then the dimensions of β will be:

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A ~~$[M^0 L T^0]$~~

B $[M L^2 T^{-2}]$

C $[M L T^{-2}]$ ^{Ans}

D ~~$[M^2 L T^2]$~~

$$M L^2 T^{-2} = \alpha \beta^2$$

$$M L^2 T^{-2} = M^{-1} T^2 \beta^2$$

$$M^2 L^2 T^{-4} = \beta^2$$

$$W = \alpha \beta^2 e^{-\frac{x^2}{\alpha kT}}$$

$$W = \alpha \beta^2 \times \frac{1}{e^{\frac{x^2}{\alpha kT}}}$$

$M^0 L^0 T^0$

$\rightarrow M^0 L^0 T^0$

$$\frac{L^2}{\alpha M L^2 T^{-2}} = M^0 L^0 T^0 \Rightarrow \alpha = M^{-1} T^2$$

QUESTION 24

$\alpha = \frac{Fv^2}{\beta^2} \log_e \left(\frac{2\pi\beta}{v^2} \right)$ (where v = velocity, F = force). Find the dimensions of α and β

A $M^0 L^2 T^{-2}, M^1 L^{-1} T^0$

B $M^1 L^1 T^{-2}, M^1 L^{-1} T^0$

C $M^{-1} L^{-1} T^0, M^0 L^2 T^2$

D $M^1 L^{-1} T^0, M^0 L^0 T^{-1}$

$\alpha = \frac{Fv^2}{\beta^2} \log \left(\frac{2\pi\beta}{v^2} \right)$

Dimensions: $[F] = MLT^{-2}$, $[v] = LT^{-1}$

Dimensions of α : $M^0 L^2 T^{-2}$

Dimensions of β : $M^1 L^{-1} T^0$

$[\alpha] = \frac{MLT^{-2} \times L^2 T^{-2}}{L^4 T^{-4}}$

$= M L^{-1} T^0$

$\frac{\beta}{(LT^{-1})^2} = M^0 L^0 T^0$

$\beta = L^2 T^{-2}$

QUESTION 25

The force of interaction between two atoms is given by $F = \alpha \beta \exp\left(-\frac{x^2}{\alpha kT}\right)$; where x is the distance, k is the Boltzmann constant and T is temperature and α and β are two constants. The dimension of β is :

JEE Main 2019 (Online) 11th January, Morning Shift

A $M^2L^2T^{-2}$

B M^2LT^{-4} *Ans*

C $M^0L^2T^{-2}$

D MLT^{-2}

$$f = \alpha \beta e^{-\frac{x^2}{\alpha kT}}$$

$$F = \alpha \beta \left[\frac{1}{e^{\frac{x^2}{\alpha kT}}} \right] M^0L^0T^0$$

$$\frac{1}{\alpha (MLT^{-2})} = 1$$

$$\alpha = M^{-1}T^2$$

$$MLT^{-2} = M^{-1}T^2 \beta$$

$$M^2LT^{-4} = \beta$$

QUESTION 27

- Physical quantity must have Units: **Yes**
- Physical Quantity must have dimensions: **No (Angle)**
- Dimensionless Physical qty can have units: **Yes (Angle)**
- All constants are dimensionless: **No**
(Gravitational Const, Planck's Const)
- Some constants may have dimensions: **Yes**
- Some constants are dimensionless: **Yes**
(1, 2, 3, 4, 5, π , e)

QUESTION 28

Match the following columns:

$f = \frac{Gm^2}{r^2}$

	Physical quantity		Dimensions		Unit
A	Gravitational constant 'G'	p	$[M^1L^1T^{-1}]$	i	<u>N m</u>
B	Torque (Work) - S - i	q	$[M^{-1}L^3T^{-2}]$	ii	N s
C	Momentum (mv) - P	r	$[M^1L^{-1}T^{-2}]$	iii	N m ² /kg ²
D	Pressure (F/A)	s	$[M^1L^2T^{-2}]$	iv	Pascal

1 A-(p)-(iii); B-(~~r~~)-(i); C-(q)-(iv); D-(r)-(ii)

2 A-(q)-(iii); B-(s)-(i); C-(p)-(ii); D-(r)-(iv)

3 A-(q)-(iii); B-(~~r~~)-(i); C-(r)-(ii); D-(s)-(iv)

4 A-(p)-(iv); B-(s)-(ii); C-(p)-(i); D-(r)-(iii)

QUESTION 29

Match List-I with List-II

[12 April, 2023 Shift-I]

List-I		List-II	
A	Spring constant MT^{-2}	I	(T^{-1})
B	Angular speed T^{-1}	II	(MT^{-2})
C	Angular momentum	III	(ML^2)
D	Moment of Inertia	IV	(ML^2T^{-1})

1 ✓ A-II, B-I, C-IV, D-III

3 A-II, B-III, C-I, D-IV

2 ✗ A-IV, B-I, C-III, D-II

4 ✗ A-I, B-III, C-II, D-IV

QUESTION 30

Match List-I with List-II

[27 Aug, 2021 Shift-II]

List-I		List-II	
A	R_H (Rydberg constant)	I	$\text{kg m}^{-1} \text{s}^{-1}$
$E=h\nu$ B	h (Planck's constant)	II	$\text{kg m}^2 \text{s}^{-1}$
C	μ_B (Magnetic field <u>energy density</u>)	III	m^{-1}
D	η (coefficient of viscosity) $F=\eta \times v$	IV	$\text{kg m}^{-1} \text{s}^{-2}$

1 A-II, B-III, C-IV, D-I**2** A-III, B-II, C-I, D-IV**3** A-III, B-II, C-IV, D-I**4** A-IV, B-II, C-I, D-III

QUESTION 31

Match List-I with List-II

[24 Jan, 2023 Shift-I]

List-I		List-II	
$E = h\nu$	A Planck's constant (h)	I	$[M^1 L^2 T^{-2}]$
$P = VI$	B Potential (V_s)	II	$[M^1 L^1 T^{-1}]$
$ML^2 T^{-2}$	C <u>Work</u> function (ϕ)	III	$[M^1 L^2 T^{-1}]$
$M \times v$	D Momentum (p)	IV	$[M^1 L^2 T^{-3} A^{-1}]$

1 A-III, B-I, C-II, D-IV**2** A-III, B-IV, C-I, D-II**3** A-II, B-IV, C-III, D-I**4** A-I, B-III, C-IV, D-II



Applications of Dimensions

Correctness of formulae

Derivation of formulae

Conversion from one System to other

$$\# T = 2\pi \sqrt{\frac{l}{g}}$$

$$D_{LHS} = T$$

$$D_{RHS} = \sqrt{\frac{L}{LT^{-2}}} = \sqrt{T^2} = T$$

$$D_{LHS} = D_{RHS} \text{ "Dimensionally Correct!"}$$

$$D_{LHS} \neq D_{RHS} \text{ Incorrect}$$

$$T = \text{Time} = [T]$$

$$l = \text{length} = [L]$$

$$g = \text{acc due to gravity} = [LT^{-2}]$$

QUESTION 32

(34) Kinetic Energy = $\left(\frac{1}{2}\right)mv^3$ → $M L^2 T^{-2}$

$$D_{LHS} = M L^2 T^{-2}$$

$$D_{RHS} = M (L T^{-1})^3$$

$$= M L^3 T^{-3}$$

$$D_{LHS} \neq D_{RHS} \quad \text{"Incorrect"}$$

(35) Velocity of wave = $\sqrt{\frac{P}{\rho}}$

$$P = \text{Pressure } M L^{-1} T^{-2}$$

$$\rho = \text{density } M L^{-3}$$

$$D_{LHS} = L T^{-1}$$

$$D_{RHS} = \sqrt{\frac{M L^{-1} T^{-2}}{M L^{-3}}} = \sqrt{L^2 T^{-2}} = L T^{-1}$$

$$D_{LHS} = D_{RHS} \quad \text{"Dim Correct"}$$

$$(33) \dot{v} = \dot{u} + \dot{a}t$$

$$D_{LHS} = LT^{-1}$$

$$D_{RHS} = LT^{-1} + LT^{-2} \times T = LT^{-1} + LT^{-1} = LT^{-1}$$

$$D_{LHS} = D_{RHS} = \text{"Dim Correct!"}$$

v = final velocity

u = initial velocity

a = acceleration

t = time

S = displacement

$$(34) S = \dot{u}t + \left(\frac{1}{2}\right)at^2$$

$$D_{LHS} = L$$

$$D_{LHS} = D_{RHS}$$

"Dim Correct"

$$D_{RHS} = L\cancel{T^{-1}} \times \cancel{T} + L\cancel{T^{-2}} \times \cancel{T^2} = L + L = L$$

$$(35) v^2 - u^2 = 2as$$

$$D_{RHS} = LT^{-2} \times L = L^2T^{-2}$$

$$D_{LHS} = (LT^{-1})^2 - (LT^{-1})^2 = L^2T^{-2} - L^2T^{-2} = L^2T^{-2}$$

$$D_{LHS} = D_{RHS} \quad \text{"Dim Correct!"}$$

(36) Velocity of wave in string = $\frac{1}{2L} \sqrt{\frac{T}{\mu}}$ ^{No.}

L = height, T = Tension, μ = Mass/Length
(Force) (ML⁻¹)

D_{LHS} = LT⁻¹

D_{RHS} = $\frac{1}{L} \sqrt{\frac{MLT^{-2}}{ML^{-1}}} = \frac{1}{L} \sqrt{L^2 T^{-2}} = \frac{1}{L} LT^{-1} = T^{-1}$

D_{LHS} ≠ D_{RHS}

"Incorrect"

(37) $P = \frac{8\eta r^4 v}{\pi v}$

P = Pressure gradient, η = Coefficient of viscosity, r = radius, v = Velocity
 L LT^{-1}

$\frac{\text{Pressure}}{\text{Length}}$

$= \frac{ML^{-1}T^{-2}}{L}$

$= ML^{-2}T^{-2}$

$F = \eta r v$

$\eta = ML^{-1}T^{-1}$

$$P = \frac{8\eta r^4 l}{\pi r}$$

$$D_{LHS} = M L^{-2} T^{-2}$$

$$D_{RHS} = \frac{M L^{-1} \cancel{T^{-1}} \times L^4 \times \cancel{L}}{\cancel{L T^{-1}}} = M L^3$$

$D_{LHS} \neq D_{RHS}$ "Incorrect".

$$(38) F = kx^3$$

$F = \text{force}, K = \text{Spring constant}, x = \text{length}$
 $MLT^{-2} \quad [MT^{-2}] \quad [L]$

$$D_{LHS} = MLT^{-2}$$

$$D_{RHS} = MT^{-2} \times L^3 = ML^3T^{-2}$$

$$D_{LHS} \neq D_{RHS} \quad \text{"Incorrect"}$$

$$(39) \sqrt{GM/r} = V$$

$F = \frac{Gm^2}{r^2}$
 $G = M^{-1}L^3T^{-2}$
 $V = \text{velocity}, G = \text{universal gravitational constant}, M = \text{Mass}, r = \text{Radius}$
 LT^{-1}

$$D_{LHS} = \sqrt{\frac{M^{-1}L^3T^{-2} \times M}{L}} \\ = \sqrt{L^2T^{-2}} = LT^{-1}$$

$$D_{RHS} = LT^{-1}$$

$$D_{LHS} = D_{RHS} \quad \text{"Dim Correct"}$$

$$(40) \lambda = \frac{h}{mv}$$

$\lambda = \text{wavelength}$, $h = \text{Planck's Constant}$, $m = \text{mass}$, $v = \text{velocity}$
 $[L]$ M LT^{-1}

$$D_{LHS} = [L]$$

$$D_{RHS} = \frac{ML^2T^{-1}}{L^3T^{-3}} = L^{-1}T^2$$

$D_{LHS} \neq D_{RHS}$ "Incorrect".

$$E = h\nu$$

$$ML^2T^{-2} = h \frac{1}{T}$$

$$ML^2T^{-1} = h$$

$$(41) F = \frac{mv^3}{r}$$

$F = \text{Force}$, $m = \text{mass}$, $v = \text{velocity}$, $r = \text{radius}$

$$D_{LHS} = MLT^{-2}$$

$D_{LHS} \neq D_{RHS}$ "Incorrect".

$$D_{RHS} = \frac{ML^3T^{-3}}{L} = ML^2T^{-3}$$

❖ Deriving a formula

Ex: Time Period of Pendulum depends on length & acc due to gravity.
derive the formula.

Sol. $T \propto l^a$
 $\propto g^b$

$$T = K l^a g^b$$

$$[M^0 L^0 T^1] = K [L]^a [L T^{-2}]^b$$

$$M^0 L^0 T^1 = K L^{a+b} T^{-2b}$$

$$a+b=0 \quad -2b=1$$

$$a = \frac{1}{2} \quad b = -\frac{1}{2}$$

$$\longrightarrow T = K l^a g^b$$

$$T = K l^{1/2} g^{-1/2}$$

$$T = K \frac{\sqrt{l}}{\sqrt{g}}$$

$$T = K \sqrt{\frac{l}{g}}$$

QUESTION 42

If P is the pressure of a gas and ρ is its density, then find the dimension of velocity

$$[ML^{-1}T^{-2}]$$

$$[ML^{-3}]$$

$$LT^{-1}$$

1 $P^{1/2}\rho^{-1/2}$ Ans

~~2 $P^{1/2}\rho^{1/2}$~~

3 $P^{-1/2}\rho^{1/2}$

~~4 $P^{-1/2}\rho^{-1/2}$~~

$$v \propto P^a$$

$$\propto \rho^b$$

$$v = K P^a \rho^b$$

$$LT^{-1} = K [ML^{-1}T^{-2}]^a [ML^{-3}]^b$$

$$M^0 L T^{-1} = K M^{a+b} L^{-a-3b} T^{-2a}$$

$$a+b=0$$

$$-a-3b=1$$

$$+2a=4$$

$$a = \frac{1}{2}$$

$$b = -\frac{1}{2}$$

$$v = K P^{1/2} \rho^{-1/2}$$

QUESTION 43

If energy (E), velocity (V) and force (F) be taken as fundamental quantity, then what are the dimensions of mass:

~~1~~ EV^2

~~3~~ FV^{-1}

2 EV^{-2}

~~4~~ FV^{-2}

$$M = K E^a V^b F^c$$

$$M^1 L^0 T^0 = K [ML^2 T^{-2}]^a [LT^{-1}]^b [MLT^{-2}]^c$$

$$M^1 L^0 T^0 = K M^{a+c} L^{2a+b+c} T^{-2a-b-2c}$$

$$a+c=1 \rightarrow a=1-c$$

$$2a+b+c=0 \rightarrow 2(1-c)+b+c=0$$

$$-2a-b-2c=0 \rightarrow 2-2c+b+c=0$$

$$\begin{cases} 2-c+b=0 \\ b-c=-2 \end{cases}$$

$$2a+b+2c=0$$

$$2(1-c)+b+2c=0$$

$$2-2c+b+2c=0$$

$$b=-2$$

$$b-c=-2$$

$$-2-c=-2$$

$$c=0$$

$$a=1$$

$$M = K E^a V^b F^c$$

QUESTION 44

If velocity $[V]$, time $[T]$ and force $[F]$ are chosen as the base quantities, the dimensions of the mass will be :

Jee Main 2021 (Online) 31st Aug. Ev. Shift

1 $[FT^{-1} V^{-1}]$

2 $[FTV^{-1}]$

3 $[FT^2 V]$

4 $[FVT^{-1}]$

$$M = K F^a T^b V^c = K F^1 T^1 V^{-1}$$

$$M^1 L^0 T^0 = K [M L T^{-2}]^a [T]^b [L T^{-1}]^c$$

$$M^1 L^0 T^0 = K M^a L^{a+c} T^{-2a+b-c}$$

$$\begin{aligned} a=1 & \quad a+c=0 & -2a+b-c=0 \\ c=-1 & & -2+b+1=0 \\ & & b=1 \end{aligned}$$

QUESTION 45

If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Youngs modulus will be:

Jee Main 2019 (Online) 11th January, Evening Shift

1 $V^{-2}A^2F^2$

2 $V^{-4}A^2F$

3 $V^{-4}A^2F$

4 $V^{-2}A^2F^{-2}$

$Y = K V^a A^b F^c$

$ML^{-1}T^{-2} = K [LT^{-1}]^a [LT^{-2}]^b [MLT^{-2}]^c$

$ML^{-1}T^{-2} = K M^c L^{a+b+c} T^{-a-2b-2c}$

$Y = \text{Pressure}$
 $= ML^{-1}T^{-2}$

$c = 1$

$a + b + c = -1$
 $a + b + 1 = -1$

$a + b = -2$

$-2b + b = -2$

$b = 2$

$-a - 2b - 2c = -2$

$a + 2b + 2c = 2$

$a + 2b + 2 = 2$

$a + 2b = 0$

$a = -2b$

$a = -4$

QUESTION 46

If force (F), length (L) and time (T) are taken as the fundamental quantities. Then what will be the dimension of density:

Jee Main 2021 (Online) 27th Aug. Ev. Shift

☒ 1 $[FL^{-4}T^2]$
Ans

☐ 2 $[FL^{-3}T^2]$

☐ 3 $[FL^{-5}T^2]$

☐ 4 $[FL^{-3}T^3]$

$$\rho = K F^a L^b T^c$$

$$ML^{-3} = K [MLT^{-2}]^a [L]^b [T]^c$$

$$ML^{-3}T^0 = K M^a L^{a+b} T^{-2a+c}$$

$$a=1$$

$$a+b=-3$$

$$1+b=-3$$

$$b=-4$$

$$-2a+c=0$$

$$-2+c=0$$

$$c=2$$

QUESTION 47

If time (t), velocity (v), and angular momentum l are taken as the fundamental units. Then the dimension of mass (m) in terms of t , v and l is :

Jee Main 2021 (Online) 20th July. Ev. Shift

1 $[t^{-1} v^1 l^{-2}]$

2 $[t^1 v^2 l^{-1}]$

3 $[t^{-2} v^{-1} l^1]$

4 $[t^{-1} v^{-2} l^1]$

$$M = K t^a v^b l^c$$

$$M^1 L^0 T^0 = K T^a [LT^{-1}]^b [ML^2 T^{-1}]^c$$

$$M^1 L^0 T^0 = K M^c L^{b+2c} T^{a-b-c}$$

$c=1$

$b+2c=0$, $a-b-c=0$

$b+2=0$

$b=-2$

$a+2-1=0$

$a+1=0$

$a=-1$

$\rightarrow \text{Mom} + \text{dis} = MLT^{-1} \times L = [ML^2 T^{-1}]$

QUESTION 48

The speed of light c , gravitational constant G and Planck's constant h are taken as fundamental units in a system. The dimensions of time in this new system should be:

(2012 Adv)

1 $[G^{1/2}h^{1/2}c^{-5/2}]$

2 $[G^{-1/2}h^{1/2}c^{-1/2}]$

3 $[G^{-1/2}h^{1/2}c^{-3/2}]$

4 $[G^{-1/2}h^{1/2}c^{1/2}]$

$$T = K G^a h^b c^c$$

$$M^0 L^0 T^1 = K [M^{-1} L^3 T^{-2}]^a [M L^2 T^{-1}]^b [L T^{-1}]^c$$

$$M^0 L^0 T^1 = K M^{-a+b} L^{3a+2b+c} T^{-2a-b-c}$$

$$-a+b=0, \quad 3a+2b+c=0, \quad -2a-b-c=1$$

$$a=b$$

$$b=\frac{1}{2}$$

$$5a+c=0$$

$$c=-5a$$

$$c=-5/2$$

$$-3a-c=1$$

$$3a+c=-1$$

$$3a-5a=-1$$

$$-2a=-1$$

$$a=1/2$$

QUESTION 49

If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is

Jee Main 2020 (Online) 2nd Sept., Ev. Shift

1 ~~$[P^2 A T^{-2}]$~~

2 ~~$[P^{\frac{1}{2}} A T^{-1}]$~~

3 $[P A^{\frac{1}{2}} T^{-1}]$

4 $[P A^{-1} T^{-2}]$

$E = K P^a A^b T^c$

$M L^2 T^{-2} = K [M L T^{-1}]^a [L^2]^b [T]^c$

$M L^2 T^{-2} = K M^a L^{a+2b} T^{-a+c}$

$a = -1$

$a + 2b = 2$

$-a + c = -2$

$-1 + c = -2$

$c = -1$

$1 + 2b = 2$

$2b = 1$

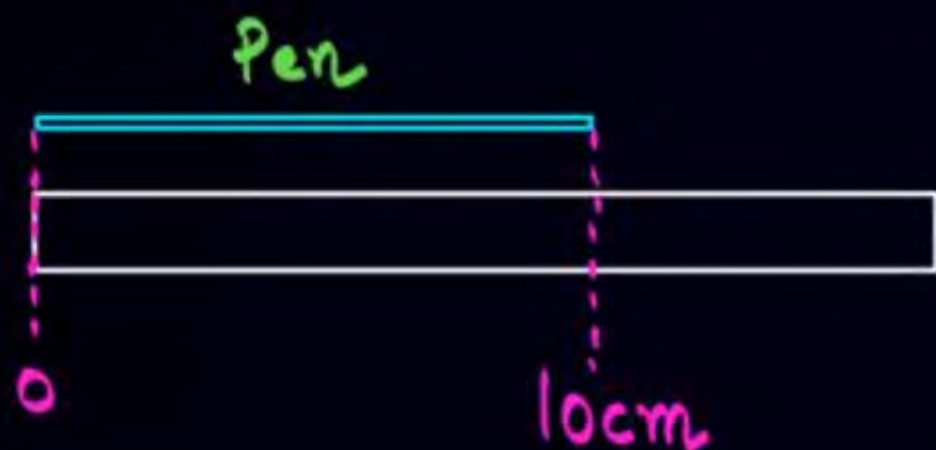
$b = \frac{1}{2}$

QUESTION 51

- Can there be a unitless quantity *Yes (Ref Index)*
- Can a unitless qty have dimensions: *No.*
- Can dimensionless Qty have units: *Yes (Angle, Solid Angle)*
- Dimensionally correct equation may be wrong: *Yes, Exp check.*
- Dimensionally incorrect equation is always wrong: *Yes.*



Conversions of Units



Pen length = 10cm

Person₁ = 10cm ✓
 Person₂ = 100mm ✓
 Person₃ = 0.1m ✓

length = 10cm { Measured → Experimentally } ⇒ (fixed)

$$n_1 u_1 = n_2 u_2$$

Conversion

○ PQ Remain Same

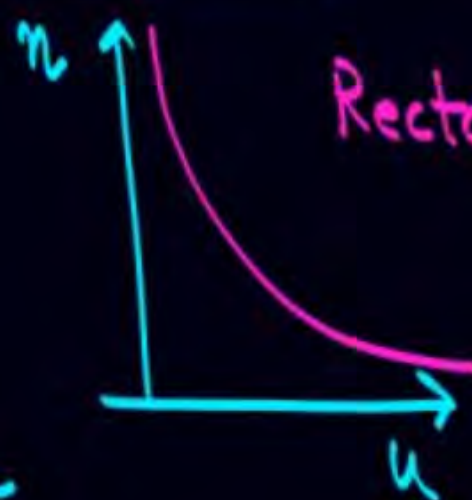
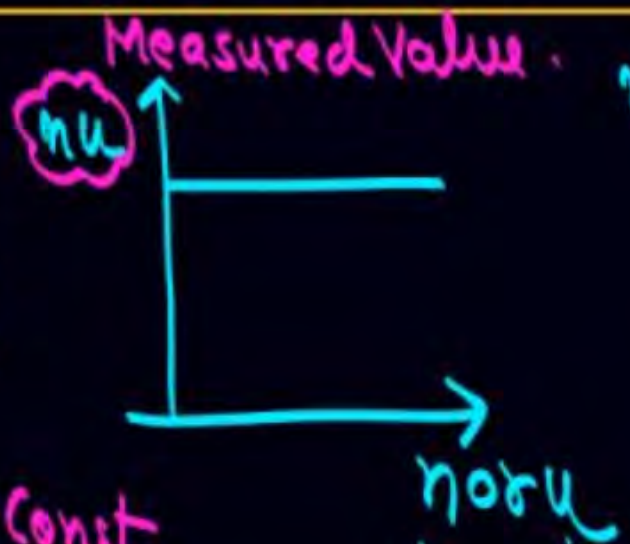
○ Measured Value is Const.

○ unit → change
 $n \rightarrow$ change

magnitude (n)
 unit (u)

$$PQ = nu = \text{Constant}$$

$$n = \frac{\text{const}}{u}$$



Rectangular Hyperbola

QUESTION 52

- Convert 36 km/hr to m/s

$$36 \text{ Km/hr} = 10 \text{ m/s}$$

$$n_1 u_1 = n_2 u_2$$

$$36 \text{ L}_1 \text{T}_1^{-1} = n_2 \text{L}_2 \text{T}_2^{-1}$$

$$\begin{aligned} n_2 &= 36 \left(\frac{\text{L}_1}{\text{L}_2} \right) \left(\frac{\text{T}_1}{\text{T}_2} \right)^{-1} = 36 \left(\frac{1000 \cancel{\text{m}}}{\cancel{\text{m}}} \right) \left(\frac{3600 \cancel{\text{s}}}{\cancel{\text{s}}} \right)^{-1} \\ &= 36 \times \frac{1000}{3600} = 10 \end{aligned}$$

- Convert 50mm² to m²

$$50 \text{ mm}^2 = 50 \times 10^{-6} \text{ m}^2$$

$$n_1 u_1 = n_2 u_2$$

$$50 \text{ L}_1^2 = n_2 \text{L}_2^2$$

$$n_2 = 50 \left[\frac{\text{L}_1}{\text{L}_2} \right]^2 = 50 \times \left[\frac{10^{-3} \cancel{\text{m}}}{\cancel{\text{m}}} \right]^2 = 50 \times 10^{-6}$$

QUESTION 53



- Convert 72000 Kg m/hr to gm cm/s

$$72,000 \text{ Kg} \frac{\text{m}}{\text{hr}} = 2 \times 10^6 \text{ gm} \frac{\text{cm}}{\text{s}}$$

$$n_1 u_1 = n_2 u_2$$

$$72000 \text{ M}_1 \text{L}_1 \text{T}_1^{-1} = n_2 \text{M}_2 \text{L}_2 \text{T}_2^{-1}$$

$$n_2 = 72,000 \left(\frac{\text{M}_1}{\text{M}_2} \right) \left(\frac{\text{L}_1}{\text{L}_2} \right) \left(\frac{\text{T}_1}{\text{T}_2} \right)^{-1}$$

$$= 72,000 \left(\frac{1000 \text{ g}}{\text{g}} \right) \left(\frac{100 \text{ cm}}{\text{m}} \right) \left(\frac{3600 \text{ s}}{\text{hr}} \right)^{-1}$$

$$= \frac{2,1000}{72,000} \times \frac{10^3 \times 10^2}{3600} = 2 \times 10^6$$

- Convert 1 m³ to cm³

$$1 \text{ m}^3 = 10^6 \text{ cm}^3$$

$$n_1 u_1 = n_2 u_2$$

$$1 \text{ L}_1^3 = n_2 \text{L}_2^3$$

$$n_2 = 1 \left[\frac{\text{L}_1}{\text{L}_2} \right]^3 = 1 \times \left[\frac{\text{m}}{10^{-2} \text{ m}} \right]^3 = 1 \times (10^2)^3 = 10^6$$

QUESTION 54



cm, gm, s.

- Convert 500 kg/m^3 of density to **CGS** System

$$500 \frac{\text{kg}}{\text{m}^3} = 0.5 \frac{\text{gm}}{\text{cm}^3}$$

$$n_1 u_1 = n_2 u_2$$

$$500 M_1 L_1^{-3} = n_2 M_2 L_2^{-3}$$

$$n_2 = 500 \left[\frac{M_1}{M_2} \right] \left[\frac{L_1}{L_2} \right]^{-3}$$

$$= 500 \left[\frac{1000 \text{g}}{\text{g}} \right] \left[\frac{100 \text{cm}}{\text{cm}} \right]^{-3}$$

$$= 5 \times 10^5 \times 10^{-6} = 0.5$$

1m = 3.2 foot

- Convert 10 m/s^2 of acceleration to **FPS** System

$$10 \text{ m/s}^2 = 32 \frac{\text{ft}}{\text{s}^2}$$

$$n_1 u_1 = n_2 u_2$$

$$10 L_1 T_1^{-2} = n_2 L_2 T_2^{-2}$$

$$n_2 = 10 \left[\frac{L_1}{L_2} \right] \left[\frac{T_1}{T_2} \right]^{-2} = 10 \left[\frac{3.2 \text{ft}}{\text{ft}} \right] \left[\frac{\text{s}}{\text{s}} \right]^{-2} = 32$$

QUESTION 55

Convert 1 Newton Of force to CGS Systems

A 10^6

B 10^5

C 10^7

D 10^9

Force (SI) $\Rightarrow mva = \text{Kg} \frac{\text{m}}{\text{s}^2}$

$$n_1 u_1 = n_2 u_2$$

$$1 \times M_1 L_1 T_1^{-2} = n_2 M_2 L_2 T_2^{-2}$$

$$n_2 = \left(\frac{M_1}{M_2} \right) \left(\frac{L_1}{L_2} \right) \left(\frac{T_1}{T_2} \right)^{-2}$$

$$n_2 = \left[\frac{1000 \text{g}}{\text{g}} \right] \left[\frac{100 \text{cm}}{\text{cm}} \right] \left[\frac{\text{s}}{\text{s}} \right]^{-2}$$

$$n_2 = 10^5$$

unit of force
in CGS.



$1\text{N} = 10^5 \text{ dynes}$

QUESTION 56

Convert 1 Joules Of work to CGS Systems

SI unit (Joules) ML^2T^{-2}

unit of work
CGS.



A 10^3

B 10^5

C 10^7

D 10^9

$$n_1 u_1 = n_2 u_2$$

$$1 \times M_1 L_1^2 T_1^{-2} = n_2 M_2 L_2^2 T_2^{-2}$$

$$n_2 = \left(\frac{M_1}{M_2} \right) \left(\frac{L_1}{L_2} \right)^2 \left(\frac{T_1}{T_2} \right)^{-2}$$

$$n_2 = \left(\frac{1000g}{g} \right) \left(\frac{100cm}{cm} \right)^2 \left(\frac{s}{s} \right)^{-2}$$

$$n_2 = 10^7$$

$$1 \text{ Joules} = 10^7 \text{ erg}$$

QUESTION 57

The density of a material in CGS system of units is 4 g/cm^3 . In a system of units in which unit of length is 10 cm and unit of mass is 100 g, the value of density of material will be: (Unit of density is Kgm^{-3})

(2011 Mains)

A 0.4

$$\frac{4 \text{ gm}}{\text{cm}^3}$$

B 0.04

$$n_1 u_1 = n_2 u_2$$

C 400

$$4 \times M_1 L_1^{-3} = n_2 M_2 L_2^{-3}$$

D 40

$$n_2 = 4 \times \left[\frac{M_1}{M_2} \right] \left[\frac{L_1}{L_2} \right]^{-3} = 4 \times \left[\frac{\text{gm}}{100 \text{ g}} \right] \left[\frac{\text{cm}}{10 \text{ cm}} \right]^{-3} = \frac{4}{100} \times 1000 = 40$$

$$L_{\text{new}} = 10 \text{ cm} = L_2$$
$$M_{\text{new}} = 100 \text{ g} = M_2$$

QUESTION 58



In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and unit of length is 10 m. In this system, one unit of power will correspond to (Unit of power is $\text{Kgm}^2\text{s}^{-3}$) $[ML^2T^{-3}]$

(2023 Mains)

$$1 \text{ Watts} = n_2 (\text{unit of Power})$$

$$n_1 u_1 = n_2 u_2$$

$$1 \times M_1 L_1^2 T_1^{-3} = n_2 M_2 L_2^2 T_2^{-3}$$

$$n_2 = \left(\frac{M_1}{M_2} \right) \left(\frac{L_1}{L_2} \right)^2 \left(\frac{T_1}{T_2} \right)^{-3} = \left(\frac{\text{kg}}{20 \text{ kg}} \right) \left(\frac{\text{m}}{10 \text{ m}} \right)^2 \left(\frac{\text{s}}{5 \text{ s}} \right)^{-3} = \frac{1}{20} \times \frac{1}{100} \times \frac{125}{1} = \frac{1}{16}$$

$$T_2 = 5 \text{ s}$$

$$M_2 = 20 \text{ kg}$$

$$L_2 = 10 \text{ m}$$

A 1/16 watts

B 16 watts

C 25 watts

D none of these

1 Watt = $\frac{1}{16}$ unit of power in new System

multiply by 16

16 Watts = 1 unit of power in new System

QUESTION 59



$$\rightarrow ML^2T^{-2} \text{ (SI)}$$

Find the value of 100 J in a system which has 20cm, 250g and half a minutes as units of length, mass and time. (Unit of Work is Kgm^2s^{-2})

(2021 Mains)

A 9×10^{-6} new units

B 2.16×10^6 new units

C 9×10^6 new units

D 100 new units

$$n_1 u_1 = n_2 u_2$$

$$100 M_1 L_1^2 T_1^{-2} = n_2 M_2 L_2^2 T_2^{-2}$$

$$n_2 = 100 \left[\frac{M_1}{M_2} \right] \left[\frac{L_1}{L_2} \right]^2 \left[\frac{T_1}{T_2} \right]^{-2}$$

$$n_2 = 100 \left[\frac{1000g}{250g} \right] \left[\frac{100cm}{20cm} \right]^2 \left[\frac{s}{30s} \right]^{-2}$$

$$= 100 \times 4 \times 25 \times 900 = 9 \times 10^6$$

$$L_2 = 20cm$$

$$M_2 = 250g$$

$$T_2 = 30s$$

$100 J = 9 \times 10^6$ (unit of work)
in new system

Ans

QUESTION 60



In a given system of units, 1 unit of mass = 2 kg, 1 unit of length = 5m and 1 unit of time = 5 sec. Then in this system, 1 N represents:

A 5/2 units of force

B 2/5 units of force

C 2 units of force

D 1/2 units of force

$$n_1 u_1 = n_2 u_2$$

$$1 \times M_1 L_1 T_1^{-2} = n_2 M_2 L_2 T_2^{-2}$$

$$n_2 = \left[\frac{M_1}{M_2} \right] \left[\frac{L_1}{L_2} \right] \left[\frac{T_1}{T_2} \right]^{-2}$$

$$n_2 = \left[\frac{1 \text{ kg}}{2 \text{ kg}} \right] \left[\frac{1 \text{ m}}{5 \text{ m}} \right] \left[\frac{1 \text{ s}}{5 \text{ s}} \right]^{-2} = \frac{1}{2} \times \frac{1}{5} \times \frac{1}{5} = \frac{1}{50}$$

$$M_2 = 2 \text{ kg}$$

$$L_2 = 5 \text{ m}$$

$$T_2 = 5 \text{ s}$$

$$1 \text{ N} = \frac{1}{50} \text{ unit of force in new system}$$

QUESTION 61

Imagine a system of units in which the unit of mass is 10 kg, length is 1 km and time is 1 minute. Then 1 J in this system is equal to 36×10^{-5} units of work:

A 360

B 3.6

C 3.6×10^5

D 36×10^{-5}

$$n_1 u_1 = n_2 u_2$$

$$1 \times M_1 L_1^2 T_1^{-2} = n_2 M_2 L_2^2 T_2^{-2}$$

$$n_2 = \left[\frac{M_1}{M_2} \right] \left[\frac{L_1}{L_2} \right]^2 \left[\frac{T_1}{T_2} \right]^{-2}$$

$$n_2 = \left(\frac{\cancel{\text{kg}}}{10 \cancel{\text{kg}}} \right) \left(\frac{\cancel{\text{m}}}{1000 \cancel{\text{m}}} \right)^2 \left(\frac{\cancel{\text{s}}}{60 \cancel{\text{s}}} \right)^{-2}$$

$$n_2 = \frac{1}{10} \times \frac{1}{1000000} \times 3600 = 36 \times 10^{-5}$$

$$M_2 = 10 \text{ Kg}$$

$$L_2 = 1000 \text{ m}$$

$$T = 60 \text{ s}$$

Drawback of dimensions:

- ① we Cannot determine Value of K .
- ① we Cannot determine whether PQ is Scalar/Vector.
- ① we Cannot determine whether formula is actually Correct.
- ① we Cannot derive formula containing $(\sin, \cos, \tan, e^x, \log x)$
- ① we Cannot derive formula like
$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as.$$

$$(Work = ML^2T^{-2})$$
$$(Torque = ML^2T^{-2})$$

ERRORS

SYSTEMATIC ERRORS

(Known / Predictable)

Where Cause is known.

Instrumental Errors

Faulty or incorrectly calibrated instruments.



Example: A scale that always shows +0.5 g.

Environmental Errors

External conditions like temperature, pressure, humidity, etc.



Example: Expansion of ruler in high temperature.

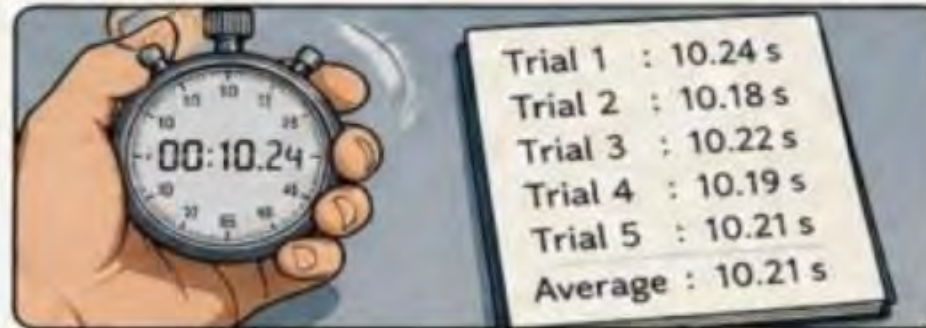
RANDOM ERRORS

(Unknown / Unpredictable)

Causes are unknown

More trials

Take multiple readings and find the average.



GROSS ERRORS

(Human Errors)

Human Errors.

Carelessness



QUICK SUMMARY

Error Type	Nature	Cause	Fix
Systematic	Predictable	Instrument/Environment	Calibration
Random	Unpredictable	Small fluctuations	Averaging
Gross	Human	Carelessness	Be careful

Random Errors → always present
Reading ↑ Random Error ↓

QUESTION 62

If random error in an experiment of 10 reading is x , how much reading we must take to reduce the random error to $x/5$

- ☐ A 20
- ☒ B 50
- ☐ C 30
- ☐ D 100

$$\rightarrow \text{Error} = \frac{1}{5}$$

Reading 5 times = 50 Readings



Representation of errors



AIM = Time period of Pendulum.

S.No	Reading
1.	1.8
2.	1.9
3.	2.0
4.	2.1
5.	2.2

No of observation = 5

- Average value / Mean Value of Time Period:

$$\bar{T} = \frac{1.8 + 1.9 + 2.0 + 2.1 + 2.2}{5} = 2s.$$

Report: Time Period of Pend. on average $= 2s = \bar{T}$

- Absolute error in readings:

Har Reading mein Kitni
galti hoti average value.

$$\Delta T_1 = 1.8 - 2.0 = -0.2$$

$$\Delta T_2 = 1.9 - 2.0 = -0.1$$

$$\Delta T_3 = 2.0 - 2.0 = 0$$

$$\Delta T_4 = 2.1 - 2.0 = +0.1$$

$$\Delta T_5 = 2.2 - 2.0 = +0.2$$

(- $\Rightarrow$ Reading smaller than
avg)

(+ $\Rightarrow$ Reading more than
avg)

- Mean Absolute error in experiment:

$$\overline{\Delta T} = \frac{|-0.2| + |-0.1| + |0| + |0.1| + |0.2|}{5} = \frac{0.6}{5} = 0.12$$

$$\text{Report: } \overline{T} \pm \overline{\Delta T} = 2 \pm 0.12$$

- Fractional error in experiment:

$$\frac{\overline{\Delta T}}{\overline{T}} = \frac{\Delta T}{T} = \frac{0.12}{2} = 0.06$$

- Percentage error in Experiment:

$$\frac{\Delta T}{T} \times 100 = 6\%$$

acceptable = 6-7%

QUESTION 63

If an experiment is performed and result of length is reported as 5 ± 0.015 .

$$\text{length} = \underline{5} \pm \underline{0.015}$$

- Average or mean value of length: 5
- Mean absolute error in length: 0.015
- Fractional error in length: $\frac{0.015}{5} = 0.003$
- Percentage error in length: $\frac{0.015}{5} \times 100 = 0.3\%$

QUESTION 64

In an experiment, refractive index of glass was observed to be 1.45, 1.56, 1.54, 1.44, 1.54 & 1.53. Calculate mean value of refractive index.

☒ **A** 1.51

Ans

6 $\rightarrow$ Readings.

RI = μ

$$\bar{\mu} = \frac{1.45 + 1.56 + 1.54 + 1.44 + 1.54 + 1.53}{6} = 1.51$$

☐ **B** 2.45

☐ **C** 1.52

☐ **D** 1.15

QUESTION 65

least Count = $L.C =$ *Sabse choti Reading Jo Measure hota.*

A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91s, 95s and 92s. If the minimum division in the measuring clock is 1s, then the reported mean time should be:

Reading = 4.

*1 → 90
2 → 91
3 → 95
4 → 92*

$$\bar{T} = 92$$

$$\begin{aligned}\Delta T_1 &= -2 \\ \Delta T_2 &= -1 \\ \Delta T_3 &= +3 \\ \Delta T_4 &= 0\end{aligned}$$

$$\bar{\Delta T} = \frac{2+1+3+0}{4} = \frac{6}{4} = 1.5.$$

$$\text{Report} = \bar{T} \pm \bar{\Delta T}$$

$$= 92 \pm 1.5s$$

$$= 92 \pm 2s \text{ Ans. } \text{Round off.}$$

A $92 \pm 1.8s$

B $92 \pm 3s$

C $92 \pm 2s$ *Ans*

D $92 \pm 5.0s$



Propagation of error

❖ Addition:

$$A = a \pm \Delta a$$

$$B = b \pm \Delta b$$

$$X = A + B$$

$$= (a+b) \pm (\Delta a + \Delta b)$$

Mean value along Se add

Mean Abs Error along Se add.

❖ Subtraction

$$X = A - B$$

$$= (a-b) \pm (\Delta a + \Delta b)$$

Mean Values Subtract

Mean Abs Error are added.

(Better analysis after differentiation in basic maths chapter)

❖ Multiplication

$$A = a \pm \Delta a$$

$$B = b \pm \Delta b$$

$$X = A \cdot B$$

fractional Error are added.

$$\frac{\Delta x}{x} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$$

❖ Division

$$X = \frac{A}{B}$$

fractional Errors are Added.

$$\frac{\Delta x}{x} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$$

QUESTION 66

Two rods have lengths measured as (1.8 ± 0.2) m and (2.3 ± 0.1) m. Calculate their combined length with error limits.

A (4.2 ± 0.3) m

☒ **B** (4.1 ± 0.3) m
Ans

C (4.1 ± 0.4) m

D (4.3 ± 0.3) m

$$d_1 = 1.8 \pm 0.2$$

$$d_2 = 2.3 \pm 0.1$$

$$d_1 + d_2 = (1.8 + 2.3) \pm (0.2 + 0.1)$$

$$= 4.1 \pm 0.3$$

QUESTION 67

Two resistors R_1 (24 ± 0.5) Ω and R_2 (8 ± 0.3) Ω are joined in series. The equivalent resistance is:

A $32 \pm 0.33 \Omega$

B $32 \pm 0.8 \Omega$

C $32 \pm 0.2 \Omega$

D $32 \pm 0.5 \Omega$

$$R_1 = 24 \pm 0.5$$

$$R_2 = 8 \pm 0.3$$

$$R_s = R_1 + R_2 = (24 + 8) \pm (0.5 + 0.3)$$
$$= 32 \pm 0.8$$

QUESTION 68

The initial and final temperatures of water as recorded by an observer are $(40.6 \pm 0.2)^\circ\text{C}$ and $(78.3 \pm 0.3)^\circ\text{C}$. Calculate the **rise in temperature** with proper error limits.

$$T_i = 40.6 \pm 0.2$$

$$T_f = 78.3 \pm 0.3$$

$$\begin{aligned}\text{Rise} &= T_f - T_i = (78.3 - 40.6) \pm (0.2 + 0.3) \\ &= 37.7 \pm 0.5\end{aligned}$$

A $37.5 \pm 0.3^\circ\text{C}$

B $37.6 \pm 0.4^\circ\text{C}$

C $37.7 \pm 0.5^\circ\text{C}$

D $37.7 \pm 0.1^\circ\text{C}$

(Better analysis after differentiation in basic maths chapter)

❖ General Case

$$X = A^n$$

$$\frac{\Delta x}{x} = n \frac{\Delta a}{a}$$

$$X = A^p B^q C^r$$

$$\frac{\Delta x}{x} = p \frac{\Delta a}{a} + q \frac{\Delta b}{b} + r \frac{\Delta c}{c}$$

$$X = \frac{A^p B^q}{C^r}$$

$$\frac{\Delta x}{x} = p \frac{\Delta a}{a} + q \frac{\Delta b}{b} + r \frac{\Delta c}{c}$$

$$K = \left(\frac{1}{2}\right) m v^2$$

Constant

$$\frac{\Delta K}{K} = \frac{\Delta m}{m} + 2 \frac{\Delta v}{v}$$

$$E = mgh$$

$$\frac{\Delta E}{E} = \frac{\Delta m}{m} + \frac{\Delta g}{g} + \frac{\Delta h}{h}$$

$$V = \sqrt{\frac{P}{\rho}} = \frac{P^{1/2}}{\rho^{1/2}}$$

$$\frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta P}{P} + \frac{1}{2} \frac{\Delta \rho}{\rho}$$

❖ General Case

(Better analysis after differentiation in basic maths chapter)

$$V = \frac{1}{2L} \sqrt{\frac{T}{\mu}} = \left(\frac{1}{2}\right) \times \frac{1}{L} \frac{T^{1/2}}{\mu^{1/2}}$$

$$\frac{\Delta V}{V} = \frac{\Delta L}{L} + \frac{1}{2} \frac{\Delta T}{T} + \frac{1}{2} \frac{\Delta \mu}{\mu}$$

$$h = \frac{2s}{r\rho g}$$

$$\frac{\Delta h}{h} = \frac{\Delta s}{s} + \frac{\Delta r}{r} + \frac{\Delta \rho}{\rho} + \frac{\Delta g}{g}$$

QUESTION 69

The resistance $R = \frac{V}{I}$, where $V = (50 \pm 2)V$ and $I = (20 \pm 0.2) A$. The percentage error in R is 'x'%. The value of 'x' to the nearest integer is_____.

Jee Main 2021 (Online) 16th March. Ev. Shift

$$R = \frac{V}{I}$$

$$V = 50 \pm 2$$
$$I = 20 \pm 0.2$$

$$R = \frac{V}{I}$$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R}{R} \times 100 = \left[\frac{\Delta V}{V} + \frac{\Delta I}{I} \right] \times 100 = \left[\frac{2}{50} + \frac{0.2}{20} \right] \times 100 = \underline{\hspace{2cm}}$$

$$\frac{\Delta R}{R} \times 100$$

QUESTION 70

The radius of a sphere is measured to be (7.50 ± 0.85) cm. Suppose the percentage error in its volume is x .

The value of x , to the nearest x , is ____.

Jee Main 2021 (Online) 18th March. Ev. Shift



$$R = 7.50 \pm 0.85$$

$$V = \left(\frac{4}{3}\pi\right) R^3$$

$$\frac{\Delta V}{V} \times 100 = x$$

$$\frac{\Delta V}{V} = 3 \frac{\Delta R}{R}$$

$$\frac{\Delta V}{V} \times 100 = 3 \times \frac{\Delta R}{R} \times 100 = 3 \times \frac{0.85}{7.5} \times 100 = \underline{\hspace{2cm}}$$

QUESTION 71

A physical quantity P is described by the relation $P = a^{1/2} b^2 c^3 d^{-4}$. If the relative errors in the measurement of a , b , c and d respectively, are 2%, 1%, 3% and 5%, then the relative error in P will be :

Jee Main 2017 (Online) 9th April Morning Shift

A 8%

B 12%

C 32%

D 25%

$$P = a^{1/2} b^2 c^3 d^{-4}$$

$$\frac{\Delta P}{P} = \frac{1}{2} \frac{\Delta a}{a} + 2 \frac{\Delta b}{b} + 3 \frac{\Delta c}{c} + 4 \frac{\Delta d}{d}$$

Multiply by 100

$$\%P = \frac{1}{2}(\%a) + 2(\%b) + 3(\%c) + 4(\%d)$$

$$= \frac{1}{2} \times 2 + 2 \times 1 + 3 \times 3 + 4 \times 5 = 1 + 2 + 9 + 20 = 32$$

$$\%a = 2$$

$$\%b = 1$$

$$\%c = 3$$

$$\%d = 5$$

QUESTION 72



What is the fractional error in g calculated from $T = 2\pi\sqrt{\ell/g}$? Given that fractional errors in T and ℓ are $\pm x$ and $\pm y$ respectively.

A $x + y$

B $x - y$

C $2x + y$ *Ans*

D $2x - y$

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T^2 = 4\pi^2 \frac{\ell}{g}$$

$$g = \frac{4\pi^2 \ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + 2\frac{\Delta T}{T} = y + 2x$$

$$\frac{\Delta T}{T} = x$$

$$\frac{\Delta \ell}{\ell} = y$$

QUESTION 73

→ fractional

$$\frac{\Delta S}{S} = \alpha$$

The relative error in the determination of the surface area of sphere is α . Then the relative error in the determination of its volume is :

Jee Main 2018 (Online) 15th April, Morning Shift

☒ **A** $\frac{3}{2}\alpha$

☐ **B** $\frac{2}{3}\alpha$

☐ **C** $\frac{5}{2}\alpha$

☐ **D** α



$$S = 4\pi R^2$$

$$\frac{\Delta S}{S} = 2 \frac{\Delta R}{R}$$

$$\alpha = 2 \frac{\Delta R}{R}$$

$$\frac{\Delta R}{R} = \frac{\alpha}{2}$$

$$V = \frac{4}{3}\pi R^3$$

$$\frac{\Delta V}{V} = 3 \frac{\Delta R}{R}$$

$$\frac{\Delta V}{V} = 3 \times \frac{\alpha}{2} = \frac{3}{2}\alpha$$

QUESTION 74



A student measures the distance traversed in free fall of a body, initially at rest in a given time. He uses this data to estimate g , the acceleration due to gravity. If the maximum percentage errors in measurement of the distance and the time are e_1 and e_2 respectively, the percentage error in the estimation of g is :-

- ☒ **A** $\frac{e_1 + 2e_2}{100}$
- ☐ **B** $e_1 + e_2$
- ☐ **C** $e_1 + 2e_2$
- ☐ **D** $e_2 - e_1$



$$T = \sqrt{\frac{2H}{g}}$$

$$T^2 = \frac{2H}{g}$$

$$g = \frac{2H}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta H}{H} + 2 \frac{\Delta T}{T}$$

$$\%g = \%H + 2\%T$$

$$= e_1 + 2e_2$$

$$\frac{\Delta H}{H} \times 100 = e_1$$

$$\frac{\Delta T}{T} \times 100 = e_2$$

QUESTION 75

A physical quantity X is given by $X = \frac{2k^3 \ell^2}{m\sqrt{n}}$. The percentage error in the measurements of k , ℓ , m and n are 1%, 2%, 3% and 4% respectively. The value of X is uncertain by

- A** 8%
- B** 10%
- C** 12%
- D** None

$$X = \frac{2K^3 \ell^2}{m n^{1/2}}$$

$$\frac{\Delta X}{X} = 3 \frac{\Delta K}{K} + 2 \frac{\Delta \ell}{\ell} + \frac{\Delta m}{m} + \frac{1}{2} \frac{\Delta n}{n}$$

$$\%X = 3(\%K) + 2(\%\ell) + (\%m) + \frac{1}{2}(\%n)$$

$$= 3(1) + 2(2) + (3) + \frac{1}{2}(4) = 3 + 4 + 3 + 2 = 12$$

QUESTION 76

A physical quantity z depends on four observables a , b , c and d , as $Z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c} d^3}$. The percentages of error in the measurement of a , b , c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is :

Jee Main 2020 (Online) 5th Sept., Morning Shift

A 13.5 %

B 14.5 %

C 16.5 %

D 12.25 %

$$Z = \frac{a^2 b^{\frac{2}{3}}}{c^{\frac{1}{2}} d^3}$$

$$\frac{\Delta Z}{Z} = 2 \frac{\Delta a}{a} + \frac{2}{3} \frac{\Delta b}{b} + \frac{1}{2} \frac{\Delta c}{c} + 3 \frac{\Delta d}{d}$$

$$\%Z = 2(\%a) + \frac{2}{3}(\%b) + \frac{1}{2}(\%c) + 3(\%d)$$

QUESTION 77

The percentage errors in quantities P, Q, R and S are 0.5%, 1%, 3% and 1.5% respectively in the measurement of a physical quantity $A = \frac{P^3 Q^2}{\sqrt{RS}}$.

The maximum percentage error in the value of A will be :

Jee Main 2018 (Online) 16th April, Morning Shift

A 6.0%

B 7.5%

C 8.5%

D 6.5%

$$A = \frac{P^3 Q^2}{R^{1/2} S^{1/2}}$$

$$\frac{\Delta A}{A} = \frac{3\Delta P}{P} + \frac{2\Delta Q}{Q} + \frac{1}{2} \frac{\Delta R}{R} + \frac{1}{2} \frac{\Delta S}{S}$$

$$\%A = 3(\%P) + 2(\%Q) + \frac{1}{2}(\%R) + \frac{1}{2}(\%S)$$

QUESTION 78

In the density measurement of a cube, the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The error in the measurement of density is: Δ

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$$m = 10 \pm 0.1 \text{ Kg}$$

$$L = 0.1 \pm 0.01 \text{ m.}$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 3 \frac{\Delta L}{L}$$

$$\rho_{\text{avg}} = \frac{m_{\text{avg}}}{L_{\text{avg}}^3} = \frac{10}{(0.1)^3} = 10 \times 10^3 = 10^4 \frac{\text{Kg}}{\text{m}^3}$$

$$\Delta \rho = \rho \left[\frac{\Delta m}{m} + 3 \frac{\Delta L}{L} \right] = 10^4 \left[\frac{0.1}{10} + 3 \frac{(0.01)}{0.1} \right] = \text{---}$$

QUESTION 80

If percentage increase in velocity is 3%. Find percentage change in Kinetic energy.

$$K = \frac{1}{2} m v^2$$

Const

$$\frac{\Delta K}{K} = 2 \frac{\Delta v}{v}$$

$$\%K = 2(\%v)$$

$$= 2 \times 3$$

$$= 6\% \text{ Ans.}$$

Agar Que mein given % Error 6-7% or less hai toh ye method laga sakte hai.

m = Constant cz not mentioned.

QUESTION 81**Error Analysis failure.**

If percentage change in velocity is 10%. Find percentage change in Kinetic energy.

Basic

Initial

$$K_i = \frac{1}{2} m v_i^2$$

$$\frac{K_f}{K_i} = 1.21$$

$$\frac{K_f}{K_i} - 1 = 1.21 - 1$$

$$\frac{K_f - K_i}{K_i} = 0.21$$

$$\frac{\Delta K}{K} = 0.21$$

$$\% = 21\%$$

Final Case

$$v_f = v_i + \frac{10}{100} v_i = 1.1 v_i$$

$$K_f = \frac{1}{2} m v_f^2$$

$$K_f = \frac{1}{2} m (1.1 v_i)^2 = 1.21 \left(\frac{1}{2} m v_i^2 \right)$$

$$K_f = 1.21 K_i$$

21%

QUESTION 82

If percentage increase in length of pendulum is 44%, Find percentage change in time period of pendulum considering gravity to be without error.

$$T_i = 2\pi \sqrt{\frac{l_i}{g}}$$

$$l_f = 1.44 l_i$$

$$T_f = 2\pi \sqrt{\frac{l_f}{g}}$$

$$T_f = 2\pi \sqrt{\frac{1.44 l_i}{g}}$$

$$= 1.2 \left(2\pi \sqrt{\frac{l_i}{g}} \right)$$

$$T_f = 1.2 T_i$$

→ 20% incr.

QUESTION 83



Two resistors of resistances $R_1 = (10 \pm 3)$ ohm and $R_2 = (20 \pm 4)$ ohm, respectively connected in parallel. Find Equivalent resistance in parallel.

$$\frac{\Delta R'}{R'^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$R_1 = 10 \pm 3$$

$$R_2 = 20 \pm 4$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_p} = \frac{1}{10} + \frac{1}{20}$$

$$R_p = \frac{10 \times 20}{30} = \frac{20}{3}$$

$$R_p = \frac{20}{3} \pm \boxed{\Delta R_p}$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\# \frac{\Delta R_p}{R_p^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\frac{\Delta R_p}{\left(\frac{20}{3}\right)^2} = \frac{3}{10^2} + \frac{4}{20^2}$$

$$\Delta R_p = \left(\frac{20}{3}\right)^2 \left[\frac{3}{100} + \frac{4}{400} \right] = \text{---}$$

QUESTION 84

$$u \rightarrow \text{Error} \rightarrow \Delta u$$
$$v \rightarrow \text{Error} \rightarrow \Delta v$$

In an experiment to measure focal length (f) of convex lens, the least counts of the measuring scales for the position of object (u) and for the position of image (v) are Δu and Δv , respectively. The error in the measurement of the focal length of the convex lens will be:

[04 April, 2024]

A $\frac{\Delta u}{u} + \frac{\Delta v}{v}$

B $f^2 \left[\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2} \right]$

C $2f \left[\frac{\Delta u}{u} + \frac{\Delta v}{v} \right]$

D $f \left[\frac{\Delta u}{u} + \frac{\Delta v}{v} \right]$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{\Delta f}{f^2} = \frac{\Delta v}{v^2} + \frac{\Delta u}{u^2}$$

$$\Delta f = f^2 \left[\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2} \right]$$

QUESTION 85

In Question if Error is not given instead L.C is given we
Can treat it as Error.

The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using wrist watch of 1 s resolution. The accuracy in the determination of g is:

Jee Main 2015 (Offline)

- A** 1%
- B** 5%
- C** 2%
- D** 3%

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T^2 = 4\pi^2 \frac{L}{g}$$

$$g = 4\pi^2 \frac{L}{T^2}$$

$$L = 20.0 \text{ cm}$$

$$\Delta L = 1 \text{ mm} = 0.1 \text{ cm}$$

$$100 \text{ s oscillation } T = 90 \rightarrow 1 \text{ oscillation } T = \frac{90}{100}$$

$$\underline{100 \text{ s}} \text{ oscillation Error} = 1 \text{ s} \rightarrow 1 \text{ oscillation Error} = \frac{1}{100}$$

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2 \frac{\Delta T}{T} \Rightarrow \frac{\Delta g}{g} \times 100 = \left[\frac{\Delta L}{L} + 2 \frac{\Delta T}{T} \right] \times 100 = \left[\frac{0.1}{20} + 2 \left(\frac{1}{90} \right) \right] \times 100$$



Significant Figures

→ whenever we take observations.
digits → Significance.

- Significant numbers: which hold Importance in many Reporting data.

Always Significant.

00^{*}20^{*}4^{*}1^{*}3^{*}9^{*}0^{*}.0^{*}2^{*}1^{*}0^{*}0^{*}

Lead
(Sig X)

Trapped.

Trail

Significant ✓

Tabhi Sig honge Jab decimal
no.

Agar decimal Nahi hai toh
Significant Nahi

- Leading Zero

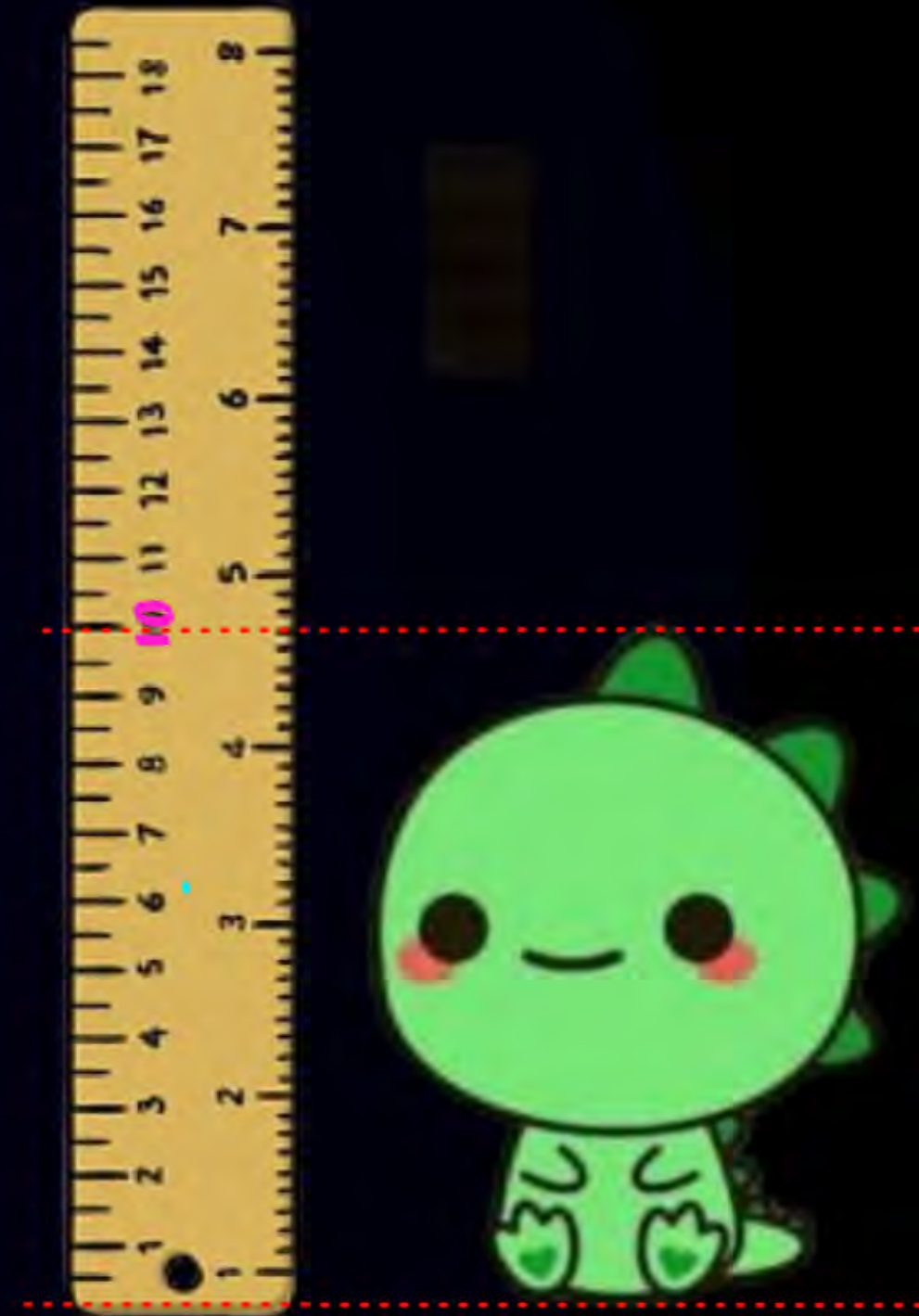
- Trailing Zero

Least count of any instrument is **uncertain**

Jab apko Error Nahi de Rakha,
toh aap L.C of Instrument ko
apna Error Maan Sakte hai!



Report = 10 cm
L.C = 0.1cm → = 10.0cm ✓
= 9.9875cm X
Error



Every measurement involves error

- So if I report my height is 1.32 mtr using scale of least count 0.01 mtr the error can be 0.01m
- The time taken by a person to move from A to B is 2.3 seconds using stop watch 0.1s the error can be 0.1s

QUESTION 86

$$\frac{\Delta Y}{Y} = \frac{\Delta M}{M} + 3\frac{\Delta L}{L} + \frac{\Delta b}{b} + 3\frac{\Delta d}{d} + \frac{\Delta \delta}{\delta}$$

A student determined Young's Modulus of elasticity using the formula $Y = \frac{MgL^3}{bd^3\delta}$. The value of g is taken to be 9.8 m/s^2 , without any significant error, his observation are as follows:

$g = \text{constant}$

[JEE Main 2021]

Then the fractional error in the measurement of Y is:

- A** 0.083
- B** 0.0155
- C** 0.0083
- D** 0.155

Physical Quantity	Least count of the equipment used for measurement	Observed value
Mass (M)	$\Delta M = 1\text{g}$	$M = 2\text{ kg}$
Length of bar (L)	$\Delta L = 1\text{ mm}$	$L = 1\text{ m}$
Breadth of bar (b)	$\Delta b = 0.1\text{ mm}$	$b = 4\text{ cm}$
Thickness of bar (d)	$\Delta d = 0.01\text{ mm}$	$d = 0.4\text{ cm}$
Depression (δ)	$\Delta \delta = 0.01\text{ mm}$	$\delta = 5\text{ mm}$

 $M = 1.000 \text{ kg. (True)}$

ACCURACY: Readings agar True Value ke pass-pass hai toh accuracy Jada hai.

PRECISION: Agar Readings as pass hai toh Precision Jada hai.

PRECISION OF INSTRUMENT: Jo Instrument Sabse choti Value Naap Sake Precision ↑

$$R_1 = 1.0 \text{ cm} \quad \Delta C = 0.1 \text{ cm}$$

$$R_2 = 1.0000 \text{ cm} \quad \Delta C = 0.0001 \text{ cm}$$

Rules for Determining Significant Digits

Rule No. 1 : All non zero digits are Significant

Rule No. 2 : All trapped zero are significant

Rule No. 3 : Leading zero are always insignificant

Rule No. 4 : Trailing zeros are significant only when decimal/bar/unit is mentioned

EX:

$3.35 \rightarrow 3$

$589 \rightarrow 3$

$283.9689 \rightarrow 7$

$23.876 \times 10^{23} \rightarrow 5$
 $\underline{\underline{=}}$ Koi load Nahi

EX:

$32003 \rightarrow 5$

$20.001 \rightarrow 5$

$101.205 \rightarrow 6$

$2.05 \times 10^{23} \rightarrow 3$

EX:

$86400 \rightarrow 3$

$240800 \rightarrow 4$

$86400. \rightarrow 5$

$240800. \rightarrow 6$

EX:

142.00 → 5

14200. → 5

14200.0 → 6

0.00256 → 3
Lead

1.00256 → 6

2.00507 → 6

4.00500 → 6
Lead → Trail
0020500 → 3

Rule No. 5 : (Conversion karte samay significant figure same rahne chahiye)

Ex:

21.3 cm SF=3

✓ SF=3 ✗ SF=6
0.213 m , 0.213000 m

✓
0.000213 km , 0.00021300
SF=3 ✗ SF=5

Rule No. 6 : Number of significant Digits does not change with scientific notation

Ex:

$\overset{\cdot\cdot}{4}\overset{\cdot\cdot}{2}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}.\text{m}$ SF = 4

In cm : $\overset{\cdot\cdot}{4}\overset{\cdot\cdot}{2}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}$, $\overset{\cdot\cdot}{4}\overset{\cdot\cdot}{2}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}$, $\overset{\cdot\cdot}{4}\overset{\cdot\cdot}{2}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0}$

$\times$ SF = 6 SF = 7

$\times$

$\overset{\cdot\cdot}{4}\overset{\cdot\cdot}{2}\overset{\cdot\cdot}{0}\overset{\cdot\cdot}{0} \times 10^5 \text{ cm}$

SF = 4

Order = 5

Scientific Notation

"a" $\times 10^b$

$1 \leq a \leq 10$

$\underline{21.3} \times 10^2 \Rightarrow 0.213 \times 10^4 \times$

$\Rightarrow \underline{2.13} \times 10^3 = \text{order}$

Order $a < 5$ Order = b

$a \geq 5$ Order = b + 1

pichla
1 2 3 (4) 5 6
aage.
drop

Round off Rules

Jis number ko drop karna hai, agar wo 5 se chota hai toh pichla number ko change nhi karna

Jis number ko drop karna hai, agar wo 5 hai toh

Jis number ko drop karna hai, agar wo 5 se bada hai toh pichla number ko +1 kar dena

Agar 5 ke baad non zero (1, 2, 3, 4, 5, 6, 7, 8, 9) koi bhi number hai toh pichle number ko +1 kar dena

Agar 5 ke baad kuch nhi ho toh pichla number agar even hai toh waise rahne do, pichla digit odd hai toh +1 kar dena

Ex:1: 25.24 round off to 3 significant Digits 25.2
... $\rightarrow$ drop < 5

Ex:2: 456.273 round off to 5 Significant Digits 456.27
... $\rightarrow$ drop < 5

Ex:3: 400.01 round off to 4 significant digits 400.0
... $\rightarrow$ drop < 5

Ex:4: 248337 round off to 3 digits 248000
... $\rightarrow$ drop

Ex:5: 25.79 round off to 3 significant Digits 25.8
... $\rightarrow$ drop > 5

Ex:6: 486.63 round off to 3 Significant Digits 487
... $\rightarrow$ drop > 5

Ex:7: 32.59 round off to 2 significant Digits 33
... $\rightarrow$ drop $= 5$

Ex:8: 3215.653 round off to 3 Significant Digits 322
... $\rightarrow$ drop $= 5$

Ex:9: 3215.653 round off to 5 Significant Digits 3215.7
... $\rightarrow$ drop $= 5$

Ex:10: 32.^{even}5 round off to 2 significant Digits **32**
...
→ drop = 5

Ex:11: 321.^{even}65 round off to 4 Significant Digits **321.6**
...
→ drop = 5

Ex:12: 32.3^{odd}5 round off to 3 significant Digits **32.4**
...
→ drop = 5

Ex:13: 4213.7^{odd}5 round off to 5 Significant Digits **4213.8**
...
↓
odd
→ drop = 5

Rules

Addition / Subtraction

Decimal utne hi rakhne hai answer mein jo, add karne wale saare numbers mein minimum decimal places ho.

SAD

Sub Add Decimal

Multiplication / Division

Answer mein utne hi significant figure rakhne hai jo minimum no of significant number ho multiply ya divide hone wale numbers ka.

MSD

Mult Div. SF

QUESTION 87

$$\begin{array}{r} \textcircled{d=3} \quad d=5 \\ 6.807 - 4.27153 = 2.535\textcircled{4}7 = 2.535 \text{ Ans} \\ \text{drop} < 5 \end{array}$$

$$\begin{array}{r} \textcircled{d=0} \quad d=1 \\ 2036 - 125.2 = 1910.\textcircled{8} = 1911 \text{ Ans} \\ \text{drop} > 5 \end{array}$$

$$\begin{array}{r} d=3 \quad d=4 \quad d=3 \quad \textcircled{d=1} \\ 2.345 + 1.8695 + 0.256 + 1.1 = 5.5\textcircled{7}05 = 5.6 \text{ Ans} \\ \text{drop} > 5 \end{array}$$

$$\begin{array}{r} d=5 \quad \textcircled{d=4} \\ 0.28956 + 2.5754 = 2.8649\textcircled{6} = 2.8650 \\ \text{drop} > 5 \end{array}$$

$$\begin{array}{r} d=5 \quad \textcircled{d=3} \\ 1.23658 - 0.135 = 1.101\textcircled{5}8 = 1.102 \text{ Ans} \\ \text{drop} = 5 \end{array}$$

$$\begin{array}{r} d=3 \quad \textcircled{d=2} \\ 5.689 - 2.63 = 3.05\textcircled{9} = 3.06 \text{ Ans} \\ \text{drop} > 5 \end{array}$$

$$\begin{array}{r} \textcircled{d=0} \quad d=4 \\ 58 - 0.0001 = 57.\textcircled{9}999 = 58 \text{ Ans} \\ \text{drop} > 5 \end{array}$$

QUESTION 88

$$\begin{array}{c} \text{S}=4 \quad \text{S}=5 \\ 2.378 \times 1.2565 = 2.987957 = 2.988 \end{array}$$

drop > 5

$$\begin{array}{c} \text{S}=4 \quad \text{①} \\ 528.2 \times 4 = 2112.8 = 2000 \end{array}$$

drop < 5

$$\begin{array}{c} \text{S}=5 \quad \text{S}=2 \\ 32.568 / 2.0 = 16.284 = 16 \end{array}$$

drop

$$\begin{array}{c} \text{S}=4 \quad \text{S}=1 \\ 1.289 \times 300 = 386.7 = 400 \end{array}$$

drop > 5

$$\begin{array}{c} \text{S}=4 \quad \text{S}=2 \\ 5.015 / 0.52 = 9.64423 = 9.6 \end{array}$$

drop < 5

$$\begin{array}{c} \text{d}=3 \quad \text{d}=1 \\ (2.023 + 1.2) \times 1.23 = (3.223) \times 1.23 = 3.96429 = 4.0 \end{array}$$

drop > 5

SF=4

Num SF hone
chahiye = 4

$$\begin{array}{c} \text{SF}=4 \\ 153.66279 / 2.5 = 61.465116 = 61 \text{ Ans} \end{array}$$

drop < 5

SF=2

Round off hamesha
Last step par
hi Karna hai.

QUESTION 89

5.74 g of a substance occupies 1.2 cm³. Express its density keeping significant figures in view.

A4.9 gm/ cm³**B**4.78 gm/ cm³**C**4.79 gm/ cm³**D**4.70 gm/ cm³

$$\rho = \frac{M}{V}$$

